

Group Projects in Partial Differential Equations (PDEs) for Geophysical Fluid Dynamics (GFD)

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Synopsis – Research in Pairs This module will comprise group projects performed with the lecturer’s help by pairs of MFC CDT students. Students, please read these notes, talk among yourselves and send me your proposed top three choices in this document for your project on the Friday before the first lecture.

Each of the ten projects outlined here is based on an Euler–Poincaré variational principle. Each project has a good chance of leading to a scientific journal publication. So, each pair of students should write up their results in a report of about ten pages of A4 L^AT_EX to be turned in at the last day of the term. These reports should be organised in order of, Why? How? What? and What next?

However, some pairs may want to get together and propose another Euler–Poincaré based GFD PDEs project. (For example, derive a variational principle for thermal geostrophy, TQG.) In that case, please write your proposal as follows: One L^AT_EX page for ‘Why?’, ‘How?’ (variational principle), ‘What you expect?’ and ‘What open problems?’ may your project may reveal.

Abstract

The family tree of GFD models is pictured in Figure 1. These equations each follow from the Euler–Poincaré theorem¹ – discussed in these notes – for a fluid Lagrangian that depends on ‘advected quantities’ such as mass, salt and heat, all carried as material properties by the fluid’s transport velocity.

Figure 2 depicts the central role played by Hamilton’s variational principle as a foundational approach in designing approximations in fluid dynamics which preserve its mathematical structure.

As shown in Figure 3, the Euler–Poincaré theorem reveals a *framework of relationships* in dynamical systems following from Lie group invariant variational principles. Figure

¹The Euler–Poincaré theorem follows for variational principles written in terms of variables that are invariant under a Lie group action. A Lie group is a group of transformations that depends smoothly on a set of parameters. For example, the Eulerian representation of fluid dynamics is written in terms of velocities that are invariant under smooth invertible maps which relabel the Lagrangian coordinates. A variational principle extremalises a time integral defined on the tangent space of a manifold. Well known examples of Lie group invariant variational principles include Fermat’s principle for geometric optics, Euler’s equations for rigid body motion, Euler’s equations for ideal fluid dynamics, and Einstein’s equations for general relativity.

³ envisions these relationships as a cube whose six faces each comprises a commuting diagram of smooth equivariant maps.

Part 1 of these notes begins by deriving the Euler–Poincaré theorem for fluid dynamics with advected quantities.² Part 2 of these notes then outlines ten group projects which demonstrate how the Euler–Poincaré theorem for fluid dynamics with advected quantities may be used to systematically formulate and derive fluid models possessing two main properties of ideal fluid dynamics – energy balance and the Kelvin circulation theorem depicted in Figure 5.³

The ten group projects described in Part 2 of these notes offer MFC CDT students the opportunity to share the experience of working together in small teams to combine and communicate their proficiencies in modelling, statistics, analysis and computational simulations to achieve a common goal, which is a mutual understanding of the mathematics underlying research in PDEs for GFD.

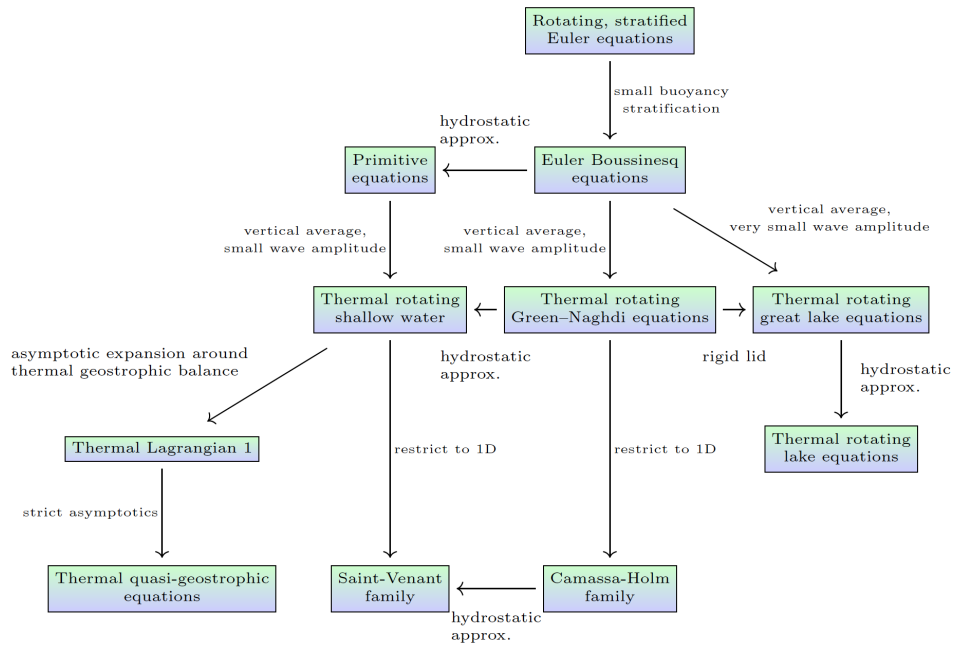


Figure 1: The family tree of GFD approximations by asymptotic expansions of Euler’s equations for a three-dimensional stratified incompressible fluid flow in a rotating frame naturally organises itself into a tree structure. All of these GFD models possess the same geometric structure – embodied in each of their Kelvin theorems – because they are each derived from approximations of the Lagrangian in Hamilton’s principle for Euler’s fluid equations [HL2021b].

²Advected quantities are pushed forward through the fixed Eulerian frame by the right action of a smooth invertible flow map, as depicted in figure 4.

³Deterministic fluid dynamics preserves both properties – energy balance and the Kelvin circulation theorem. However, in the stochastic case, the Euler–Poincaré framework forces a choice of either conserved deterministic energy, or a stochastic Kelvin circulation theorem with noisy transport velocity.

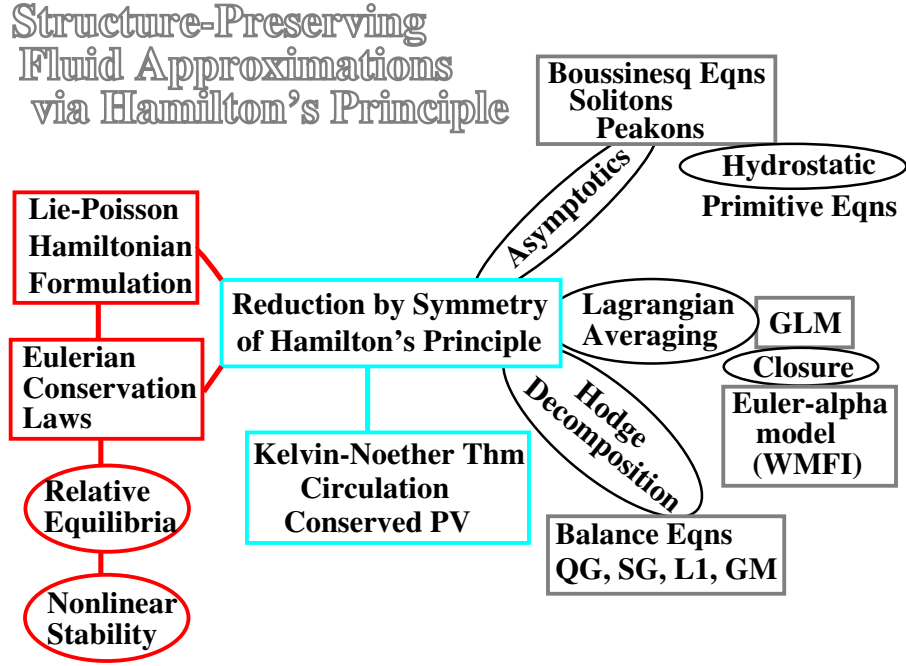


Figure 2: Averaging, asymptotic expansions and other approximations of the Lagrangian in Hamilton's principle provide a cascade of structure preserving approximations of fluid dynamics, all of which retain the fundamental properties of the Kelvin circulation theorem and conservation laws associated with potential vorticity. They also all retain an extremely useful shared Hamiltonian formulation [MPEbook2017].

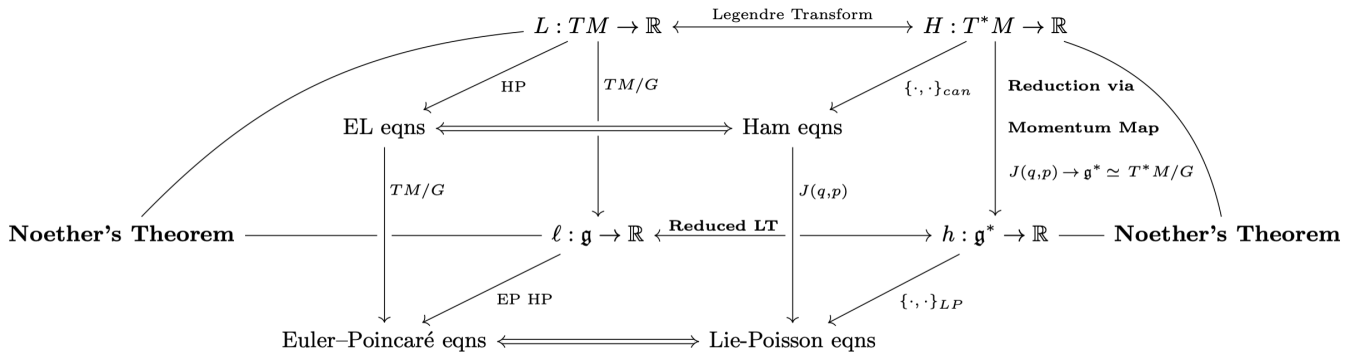


Figure 3: The Euler-Poincaré theorem reveals a *framework of relationships* in dynamical systems following from Lie group invariant variational principles. These relationships are shown here as a cube whose six faces each comprises a commuting diagram of smooth equivariant maps [H2025].

$$\begin{array}{ccc}
 & T^* \text{Emb}(S, M) & \\
 \mathbf{J}_{Sing} \swarrow & & \searrow \mathbf{J}_S \\
 \mathfrak{X}(M)^* & & \mathfrak{X}(S)^*
 \end{array}$$

Figure 4: The figure represents the dual pair J_S and J_{Sing} of momentum maps for continuum mechanics [HM2005]. On the right, J_S is Eulerian (i.e., taking time-dependent values at fixed points in the lab frame and right invariant under pullback by the flow). On the left, J_{Sing} is Convective (i.e., embedded in the flow, and left invariant under push-forward by the flow).

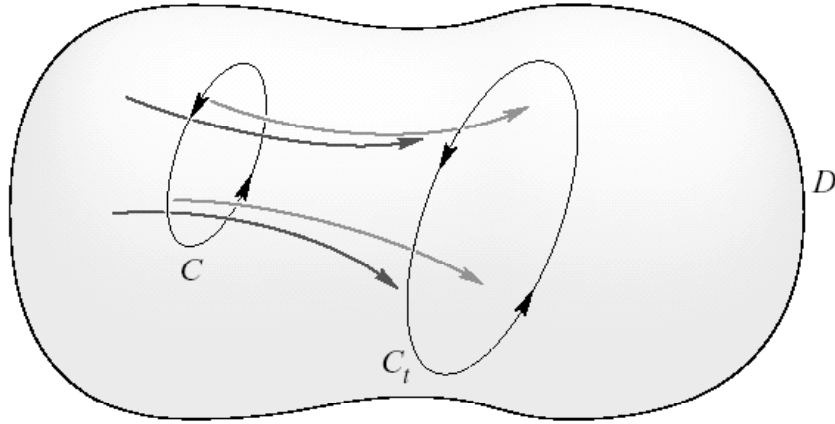


Figure 5: The Kelvin circulation theorem represents Newton's law for a material loop C advecting under a flow map ϕ_t by $C_t = \phi_t \circ C$ at Eulerian velocity $u := \dot{\phi}_t \phi_t^{-1}$ with $\phi_0 = Id$. The Kelvin form of Newton's force law (force is time derivative of momentum) is rephrased in fluid terms by writing $dP/dt = F$ in the Euler–Poincaré form as a distribution of the fluid mass on a material loop moving with the flow ϕ_t [MPEbook2017, H2025].

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1 Preparation for applying Hamilton's principle in GFD

Contents

1.1	Defining variational derivatives	6
1.2	What is advection, mathematically?	7
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1.1 Defining variational derivatives

Definition

1.1. The **variational derivative** $\frac{\delta F}{\delta \phi} \in V^*(M)$ of a real-valued functional $F[\phi]$ of smooth functions $\phi \in \mathcal{F}^\infty(M)$ taking values in a vector space $V(M)$ over a manifold $M \subset \mathbb{R}^n$ ⁴ is defined by the Frechet derivative

$$\delta F[\phi] = \lim_{\varepsilon \rightarrow 0} \left(F[\phi + \varepsilon \delta \phi] - F[\phi] \right) = \left\langle \frac{\delta F}{\delta \phi}, \delta \phi \right\rangle \quad \text{for} \quad \delta \phi = \lim_{\varepsilon \rightarrow 0} \partial_\varepsilon \phi(\mathbf{x}, t, \varepsilon) \in \mathcal{F}^\infty(M). \quad (1.1)$$

The angle brackets $\langle \cdot, \cdot \rangle$ here denote L^2 pairing, as in

$$\left\langle \frac{\delta F}{\delta \phi}, \delta \phi \right\rangle_{V^* \times V} = \int_M \left(\frac{\delta F}{\delta \phi}(\mathbf{x}, t) \delta \phi(\mathbf{x}, t) \right) d^n x, \quad (1.2)$$

for integrable real functions $\phi \in V$ and $\frac{\delta F}{\delta \phi} \in V^*$, and pairing $\langle \cdot, \cdot \rangle_{V^* \times V}: V^* \times V \rightarrow \mathbb{R}$, for a vector space V and its dual vector space V^* .

This definition of variational derivative applies in the present case for functionals

$$F(\tilde{u}, \tilde{D}, b), \quad u := \mathbf{u} \cdot \nabla \in \mathfrak{X}(\mathbb{R}^n) \times (\tilde{D} := D d^n x) \in Den(\mathbb{R}^n) \times \phi \in \mathcal{F}^\infty(\mathbb{R}^n) \rightarrow \mathbb{R}, \quad (1.3)$$

since smooth vector fields $\mathfrak{X}(\mathbb{R}^n)$, densities $Den(\mathbb{R}^n)$ and functions $\mathcal{F}^\infty(\mathbb{R}^n)$ are all vector spaces. In L^2 pairing densities are dual to functions and (contravariant) vector fields are dual to (co-variant) 1-form densities.

References

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⁴A manifold is a space where the rules of calculus apply; in particular, the product rule and the chain rule.

1.2 What is advection, mathematically?

Definition

1.2 (An advected quantity is invariant along a flow trajectory).

When the **pull-back relation** is applied to this definition for $x_t = \phi_t(x_0)$

$$\alpha_0(x_0) = \alpha_t(x_t) = (\phi_t^* \alpha_t)(x_0),$$

one finds that advected quantities satisfy the **transport formula**,

$$0 = \frac{d}{dt} \alpha_0(x_0) = \frac{d}{dt} (\phi_t^* \alpha_t)(x_0) = \phi_t^* (\partial_t + \mathcal{L}_u) \alpha_t(x_0) = (\partial_t + \mathcal{L}_u) \alpha_t(x_t),$$

where the vector field $u = \dot{\phi}_t \phi_t^{-1}$ generates the flow map ϕ_t . The last statement is clear because of the definition of Eulerian velocity in terms of Lagrangian velocity:

$$\dot{x}_t = \dot{\phi}_t(x_0) =: u_t(x_t) = u_t(\phi_t(x_0)) = \phi_t^* u_t(x_0) \implies u_t = \dot{\phi}_t \phi_t^{-1}.$$

Equivalently, via the **push-forward relation**,

$$\alpha_t(x_t) = (\alpha_0 \circ \phi_t^{-1})(x_t) = ((\phi_t)_* \alpha_0)(x_t),$$

implies the **same transport formula**,

$$\frac{d}{dt} \alpha_t(x_t) = \frac{d}{dt} (\phi_t)_* \alpha_0 = -(\mathcal{L}_X \alpha_t)(x_t).$$

Example

1.3 (Familiar examples of Lie derivatives from fluid dynamics in three dimensions).⁵

(Functions) $(\partial_t + \mathcal{L}_u)b(\mathbf{x}, t) = \partial_t b + \mathbf{u} \cdot \nabla b$ for a vector field $u = \dot{g}_t g_t^{-1}$,

(Vector fields) $(\partial_t + \mathcal{L}_u)w(\mathbf{x}, t) = \partial_t w - \text{ad}_u w = \partial_t w - [u, w] = (\partial_t \mathbf{w} + \mathbf{u} \cdot \nabla \mathbf{w} - \mathbf{w} \cdot \nabla \mathbf{u}) \cdot \nabla$

(1-forms) $(\partial_t + \mathcal{L}_u)(\mathbf{v}(\mathbf{x}, t) \cdot d\mathbf{x}) = (\partial_t \mathbf{v} + \mathbf{u} \cdot \nabla \mathbf{v} + v_j \nabla u^j) \cdot d\mathbf{x}$
 $= (\partial_t \mathbf{v} - \mathbf{u} \times \text{curl } \mathbf{v} + \nabla(\mathbf{u} \cdot \mathbf{v})) \cdot d\mathbf{x},$

(2-forms) $(\partial_t + \mathcal{L}_u)(\boldsymbol{\omega}(\mathbf{x}, t) \cdot d\mathbf{S}) = (\partial_t \boldsymbol{\omega} - \text{curl}(\mathbf{u} \times \boldsymbol{\omega}) + \mathbf{u} \text{div } \boldsymbol{\omega}) \cdot d\mathbf{S},$

(3-forms) $(\partial_t + \mathcal{L}_u)(D(\mathbf{x}, t) d^3x) = (\partial_t D + \text{div } D \mathbf{u}) d^3x.$

1.3 Variational formulae in three dimensions

In fluid dynamical applications, the advected Eulerian variables b and $D d^3x$ represent the buoyancy b (or specific entropy, for the compressible case) and volume element (or mass density) $D d^3x$, respectively. In the Euler–Poincaré theory, the variations of the tensor functions a at fixed $\mathbf{x}$ and t are also given by advective Lie derivatives, namely $\delta a = -\mathcal{L}_w a$, for a vector field $w = \delta g_\epsilon g_\epsilon^{-1}$,

$$\begin{aligned} \delta b &= -\mathcal{L}_w b = -\mathbf{w} \cdot \nabla b, \text{ for a vector field } w = \partial_\epsilon g_\epsilon g_\epsilon^{-1} \Big|_{\epsilon=0} \\ \delta(D d^3x) &= -\mathcal{L}_w (D d^3x) = -\nabla \cdot (D \mathbf{w}) d^3x, \\ \delta u &= \delta(\mathbf{u} \cdot \nabla) = \partial_t w - \text{ad}_u w = (\partial_t \mathbf{w} + \mathbf{u} \cdot \nabla \mathbf{w} - \mathbf{w} \cdot \nabla \mathbf{u}) \cdot \nabla. \end{aligned} \tag{1.4}$$

⁵Here we denote the flow map as g_t because the flow map is assumed to satisfy the group properties under composition of maps: $\phi_s \circ \phi_t = \phi_{t+s}$.

Hence, Hamilton's principle with this dependence and these variations yields

$$\begin{aligned}
0 &= \delta \int_{\mathcal{D}} \ell(u, b, D) dt \\
&= \int_{\mathcal{D}} \left\langle \frac{\delta \ell}{\delta u}, \delta u \right\rangle + \left\langle \frac{\delta \ell}{\delta b}, \delta b \right\rangle + \left\langle \frac{\delta \ell}{\delta D}, \delta D \right\rangle dt \\
&= \int_{\mathcal{D}} \left[\frac{\delta \ell}{\delta \mathbf{u}} \cdot \left(\frac{\partial \mathbf{w}}{\partial t} - \text{ad}_{\mathbf{u}} \mathbf{w} \right) - \frac{\delta \ell}{\delta b} \mathbf{w} \cdot \nabla b - \frac{\delta \ell}{\delta D} \left(\nabla \cdot (D \mathbf{w}) \right) \right] d^3 x dt \\
&= \int_{\mathcal{D}} \mathbf{w} \cdot \left[- \frac{\partial}{\partial t} \frac{\delta \ell}{\delta \mathbf{u}} - \text{ad}_{\mathbf{u}}^* \frac{\delta \ell}{\delta \mathbf{u}} - \frac{\delta \ell}{\delta b} \nabla b + D \nabla \frac{\delta \ell}{\delta D} \right] d^3 x dt \\
&= - \int_{\mathcal{D}} \mathbf{w} \cdot \left[\left(\frac{\partial}{\partial t} + \mathcal{L}_u \right) \frac{\delta \ell}{\delta \mathbf{u}} + \frac{\delta \ell}{\delta b} \nabla b - D \nabla \frac{\delta \ell}{\delta D} \right] d^3 x dt \\
&= - \int_{\mathcal{D}} \mathbf{w} \cdot \left[\left(\frac{\partial}{\partial t} + \mathcal{L}_u \right) \frac{1}{D} \frac{\delta \ell}{\delta \mathbf{u}} + \frac{1}{D} \frac{\delta \ell}{\delta b} \nabla b - \nabla \frac{\delta \ell}{\delta D} \right] D d^3 x dt
\end{aligned} \tag{1.5}$$

where in the last step we have used the auxiliary Lie-derivative relation for the continuity equation $(\partial/\partial t + \mathcal{L}_u)Dd^3x = 0$ to factor out Dd^3x . We have also dropped boundary terms arising from integrations by parts, by invoking ‘natural boundary conditions’. Specifically, we have imposed $\hat{\mathbf{n}} \cdot \mathbf{w} = 0$ on the boundary, where $\hat{\mathbf{n}}$ is the boundary's outward unit normal vector and $w = \delta g_t \circ g_t^{-1}$ vanishes at the endpoints in time.

1.4 Euler–Poincaré framework for geophysical fluid dynamics (GFD)

The Euler–Poincaré equations for continua may now be summarized in vector form for advected Eulerian variables a in the example set 1.3. The last step in the calculation (1.5) expresses the Euclidean components of the Euler–Poincaré equations for continua in Kelvin theorem form

$$\left(\frac{\partial}{\partial t} + \mathcal{L}_u \right) \left(\frac{1}{D} \frac{\delta \ell}{\delta \mathbf{u}} \cdot d\mathbf{x} \right) + \frac{1}{D} \frac{\delta \ell}{\delta b} \nabla b \cdot d\mathbf{x} - \nabla \left(\frac{\delta \ell}{\delta D} \right) \cdot d\mathbf{x} = 0, \tag{1.6}$$

in which the variational derivatives of the Lagrangian $\ell(\mathbf{u}, b, D)$ are computed as Fréchet derivatives, according to the usual physical conventions. Formula (1.6) is the Kelvin–Noether form of the equation of motion for ideal continua. Hence, we have the explicit Kelvin theorem expression,

$$\frac{d}{dt} \oint_{c_t(\mathbf{u})} \frac{1}{D} \frac{\delta \ell}{\delta \mathbf{u}} \cdot d\mathbf{x} = - \oint_{c_t(\mathbf{u})} \frac{1}{D} \frac{\delta \ell}{\delta b} \nabla b \cdot d\mathbf{x}, \tag{1.7}$$

where the curve $c_t(\mathbf{u})$ moves with the fluid velocity $\mathbf{u}$. Then, by Stokes' theorem, the Euler equations generate circulation of $\mathbf{v} := (D^{-1} \delta \ell / \delta \mathbf{u})$ whenever the gradients ∇b and $\nabla(D^{-1} \delta \ell / \delta b)$ are not collinear. The corresponding *conservation of potential vorticity* q on fluid parcels is given by

$$\frac{\partial q}{\partial t} + \mathbf{u} \cdot \nabla q = 0, \quad \text{where} \quad q = \frac{1}{D} \nabla b \cdot \text{curl} \left(\frac{1}{D} \frac{\delta \ell}{\delta \mathbf{u}} \right). \tag{1.8}$$

This is also called *PV convection*. Equations (1.6)–(1.8) embody most of the panoply of equations for GFD. The vector form of equation (1.6) is,

$$\underbrace{\left(\frac{\partial}{\partial t} + \mathbf{u} \cdot \nabla \right) \left(\frac{1}{D} \frac{\delta \ell}{\delta \mathbf{u}} \right) + \frac{1}{D} \frac{\delta \ell}{\delta u^j} \nabla u^j}_{\text{Geodesic Nonlinearity: Kinetic energy}} = \underbrace{\nabla \frac{\delta \ell}{\delta D} - \frac{1}{D} \frac{\delta \ell}{\delta b} \nabla b}_{\text{Potential energy}} \tag{1.9}$$

In geophysical applications, the Eulerian variable D represents the frozen-in volume element and b is the buoyancy. In this case, *Kelvin's theorem* is

$$\frac{dI}{dt} = \iint_{S(t)} \nabla \left(\frac{1}{D} \frac{\delta l}{\delta b} \right) \times \nabla b \cdot d\mathbf{S},$$

with circulation integral

$$I = \oint_{b(t)} \frac{1}{D} \frac{\delta l}{\delta \mathbf{u}} \cdot d\mathbf{x}.$$

An alternative vector form of equation (1.9) is,

$$\frac{\partial}{\partial t} \left(\frac{1}{D} \frac{\delta l}{\delta \mathbf{u}} \right) - \mathbf{u} \times \nabla \text{curl} \left(\frac{1}{D} \frac{\delta l}{\delta \mathbf{u}} \right) + \nabla \left(\mathbf{u} \cdot \frac{1}{D} \frac{\delta l}{\delta \mathbf{u}} \right) = \nabla \frac{\delta l}{\delta D} - \frac{1}{D} \frac{\delta l}{\delta b} \nabla b. \quad (1.10)$$

1.5 The general case of the Euler-Poincaré equations for continua

To express the general case for applications of the Euler-Poincaré equations for continua, we need a general definition for the terms on the right-hand side of these equations. These terms are called ‘the diamond terms’. The diamond terms are what gives the Euler-Poincaré equation its general applicability.

Definition

1.4 (The ‘diamond’ operation). The operation $\diamond : V \times V^* \rightarrow \mathfrak{X}^*$ between tensor space elements $a \in V^*$ and $b \in V$ produces an element of $\mathfrak{X}(\mathcal{D})^*$, a one-form density, defined by

$$\langle b \diamond a, u \rangle_{\mathfrak{X}} = - \int_{\mathcal{D}} b \cdot \mathcal{L}_u a =: \langle b, -\mathcal{L}_u a \rangle_V,$$

where $\langle \cdot, \cdot \rangle_{\mathfrak{X}}$ denotes the symmetric, non-degenerate L^2 pairing between vector fields and one-form densities, which are dual with respect to this pairing. Likewise, $\langle \cdot, \cdot \rangle_V$ represents the corresponding L^2 pairing between dual elements of V and V^* .

Also, $\mathcal{L}_u a$ stands for the Lie derivative of an element $a \in V^*$ with respect to a vector field $u \in \mathfrak{X}(\mathcal{D})$, and $b \cdot \mathcal{L}_u a$ denotes the contraction between elements $b \in V$ and elements $a \in V^*$.

Here are some examples which explain what diamond ($\diamond$) means operationally for fluid dynamics:

Example

1.5 (Calculating diamond ($\diamond$) with vector calculus). In $\mathbb{R}^3$, for a denoted variously as $\theta \in \Lambda^0$, $A = \mathbf{A} \cdot d\mathbf{x} \in \Lambda^1$, $B = \mathbf{B} \cdot d\mathbf{S} \in \Lambda^2$, $D = D d^3x \in \Lambda^3$ and $S = S_{ab} dx^a \otimes dx^b \in T_2^0(\text{sym})$

$$\begin{aligned} \left(\frac{\delta l}{\delta a} \diamond a \right) &= \left[- \frac{\delta l}{\delta \theta} \nabla \theta \cdot d\mathbf{x} + D \nabla \left(\frac{\delta l}{\delta D} \right) \cdot d\mathbf{x} \right. \\ &\quad + \left(\frac{\delta l}{\delta \mathbf{A}} \times \text{curl } \mathbf{A} - \mathbf{A} \text{div} \frac{\delta l}{\delta \mathbf{A}} \right) \cdot d\mathbf{x} \\ &\quad + \left(\text{curl} \frac{\delta l}{\delta \mathbf{B}} \times \mathbf{B} - \frac{\delta l}{\delta \mathbf{B}} \text{div} \mathbf{B} \right) \cdot d\mathbf{x} \\ &\quad \left. + \left(\frac{\delta l}{\delta S_{ab}} S_{ab,k} - \left(\frac{\delta l}{\delta S_{ab}} S_{kb} \right)_{,a} - \left(\frac{\delta l}{\delta S_{ab}} S_{ka} \right)_{,b} \right) dx^k \right] \otimes d^3x. \end{aligned}$$

Now we are in a position to state the general case of the Euler-Poincaré equations and its Kelvin-Noether Circulation Theorem.

Theorem

1.6 (Kelvin-Noether Circulation Theorem). Assume that the vector field $u(x, t)$ satisfies the Euler-Poincaré equations for continua:

$$\left(\frac{\partial}{\partial t} + \mathcal{L}_u \right) \left(\frac{\delta l}{\delta u} \right) = \frac{\delta l}{\delta a} \diamond a$$

and the advected quantity a satisfies the Lie derivative relation

$$\frac{\partial a}{\partial t} + \mathcal{L}_u a = 0.$$

Let g_t be the flow of the Eulerian velocity field u , that is, $u_t = (dg_t/dt) \circ g_t^{-1}$. Then for an initial material loop γ_0 one has the flow trajectory, $\gamma_t := g_t \circ \gamma_0$ and $I(t) := I(\gamma_t, u_t, a_t)$. For these conditions

$$\frac{d}{dt} I(t) = \oint_{\gamma_t} \frac{1}{D} \frac{\delta l}{\delta a} \diamond a.$$

Remark

1.7. In the statement of the Kelvin-Noether theorem, we use a subscript t to emphasise that the operations are applied at a particular time t and to avoid having to write the other arguments, as in $a_t(x) = a(x, t)$. That is, we omit the arguments from the notation when convenient. Due to the importance of this theorem we shall provide a separate proof tailored for the case of continuum mechanical systems.

Proof. First we change variables in the expression for the circulation integral $I(t)$:

$$I(t) = \oint_{\gamma_t} \frac{1}{D_t} \frac{\delta l}{\delta u} = \oint_{\gamma_0} g_t^* \left[\frac{1}{D_t} \frac{\delta l}{\delta u} \right] = \oint_{\gamma_0} \frac{1}{D_0} g_t^* \left[\frac{\delta l}{\delta u} \right].$$

Next, we use the Lie derivative formula

$$\frac{d}{dt} (g_t^* \alpha_t) = g_t^* \left(\frac{\partial}{\partial t} \alpha_t + \mathcal{L}_v \alpha_t \right)$$

for an arbitrary one-form density α_t . This formula gives

$$\begin{aligned} \frac{d}{dt} I(t) &= \frac{d}{dt} \oint_{\gamma_0} \frac{1}{D_0} g_t^* \left[\frac{\delta l}{\delta u} \right] \\ &= \oint_{\gamma_0} \frac{1}{D_0} \frac{d}{dt} \left(g_t^* \left[\frac{\delta l}{\delta u} \right] \right) \\ &= \oint_{\gamma_0} \frac{1}{D_0} g_t^* \left[\frac{\partial}{\partial t} \left(\frac{\delta l}{\delta u} \right) + \mathcal{L}_u \left(\frac{\delta l}{\delta u} \right) \right]. \end{aligned}$$

By the Euler-Poincaré equations, this becomes

$$\frac{d}{dt} I(t) = \oint_{\gamma_0} \frac{1}{D_0} g_t^* \left[\frac{\delta l}{\delta a} \diamond a \right] = \oint_{\gamma_t} \frac{1}{D_t} \left[\frac{\delta l}{\delta a} \diamond a \right],$$

once again by the change of variables formula.

Since the last expression holds for every loop γ_t , we may write it as

$$\left(\frac{\partial}{\partial t} + \mathcal{L}_u\right) \frac{1}{D} \frac{\delta l}{\delta u} = \frac{1}{D} \frac{\delta l}{\delta a} \diamond a.$$

This is the Kelvin-Noether form of the Euler-Poincaré equations for ideal continuum dynamics. $\square$

For a good introduction to geometric mechanics for field theories on YouTube, see [L25]. For a textbook about geometric mechanics leading from finite dimensions to continuum field theories, see [HSS2009].

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Youtube.com + search(Introduction to geometric mechanics)

2 Group project topics

What is this part about? This part outlines ten topics for group projects assessing progress in learning GFD research methods. These ten topics are designed to emphasise modelling, statistics and analysis; sometimes deterministic and sometimes stochastic. All of the discussion in Part 2 is closely related to Part 1.

2.1 Creation of rogue waves by bathymetry: Investigated via an Euler–Poincaré principle based on information transport

Can we really deal with rogue waves?



Figure 6: No. We are unable to use diffeomorphisms when wave breaking occurs.

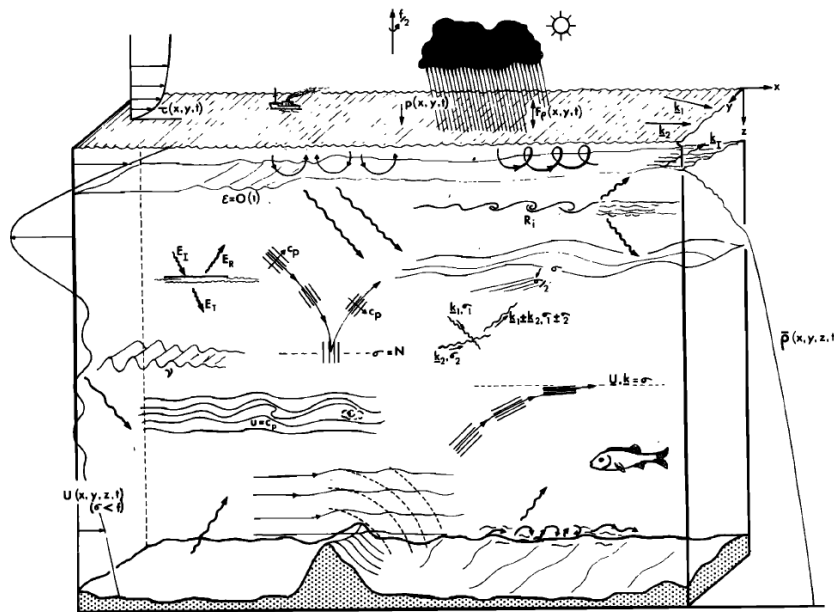


Fig. 5. Physical processes affecting internal waves.

Figure 7: We deal with continuum flows of currents & propagating waves acting back on them. Its multi-scale, multi-physics flow dynamics can be modelled in the Euler–Poincaré framework.

What type of statistics is needed to deal with rogue waves?

What kind of statistics be combined with fluid flow? The Fisher–Rao metric on the space $\text{Prob}(M)$ of probability densities is of importance in the field of information geometry. For a review of its usage as a gradient flow of information in finite dimensions, see [Towell2025]. Instead of a gradient flow, though, this project intends to describe *information transport* in infinite dimensions as a type of continuum flow in the framework of Euler–Poincaré variational dynamics based in Hamilton’s principle with a Lagrangian that is invariant under transformations by smooth invertible maps (diffeomorphism). For this purpose, we must choose the Fisher–Rao metric; because the Fisher–Rao metric is unique in its invariance under diffeomorphisms acting from the right on the space $\text{Prob}(M)$ of probability densities.

The Fisher–Rao metric on the infinite-dimensional manifold of all positive probability densities was studied in [Cencov1982], including the computation of its curvature.

A recent theorem proved in [BBM2016] extended to infinite-dimensions a result for the Fisher–Rao metric in finite dimensions proved in [Friedrich1991], as follows.

Theorem

2.1 (Bauer, Bruveris and Michor 2016). *Let M be a compact manifold without boundary of dimension ≥ 2 . Then any smooth weak Riemannian metric on the space $\text{Prob}(M)$ of smooth positive probability densities that is invariant under the action of the diffeomorphism group of M , is a multiple of the Fisher–Rao metric.*

Defining the Fisher–Rao metric. Following [BBM2016], let M^m be a smooth compact manifold without boundary. Denote by $\text{Dens}_+(M)$ the space of smooth positive densities on M , i.e. $\text{Dens}_+(M) = \{D \in \Gamma(\text{Vol}(M)) : D(x) > 0 \text{ for all } x \in M\}$. Let $\text{Prob}(M)$ be the subspace of positive densities which integrate to 1 on M . For $D \in \text{Dens}_+(M)$ the tangent spaces is $T_D \text{Dens}_+(M) = \Gamma(\text{Vol}(M))$ and for $D \in \text{Prob}(M)$ one has

$$T_D \text{Prob}(M) = \left\{ \alpha \in \Gamma(\text{Vol}(M)) : \int_M \alpha = 0 \right\}.$$

The Fisher–Rao metric is a Riemannian metric on $\text{Prob}(M)$ defined as

$$G_D^{\text{FR}}(\alpha, \beta) := \int_M \frac{\alpha}{D} \frac{\beta}{D} D.$$

This metric is right invariant under the associated action of $\text{Diff}(M)$ on $\text{Prob}(M)$, since

$$\varphi^* \left(G_D^{\text{FR}}(\alpha, \beta) \right) = G_{\varphi^* D}^{\text{FR}}(\varphi^* \alpha, \varphi^* \beta) = \int_M \left(\frac{\alpha}{D} \circ \varphi \right) \left(\frac{\beta}{D} \circ \varphi \right) \varphi^* D = \int_M \frac{\alpha}{D} \frac{\beta}{D} D,$$

where φ^* denotes pull-back, i.e., acting from the right with the smooth invertible map $\varphi \in \text{Diff}(\text{Prob}(M))$.

Exercise. Consider the constrained Hamilton’s principle,

$$\begin{aligned} S &= \int_0^T l(\mathbf{u}, D) dt = \int_0^T \int_D \frac{1}{2} D |\mathbf{u}|^2 - \frac{d^2 |\nabla D|^2}{2 D^2} D - \frac{D^2 - 2DD_0}{2\mathcal{F}} \\ &\quad - \phi(\partial_t D + \text{div}(D\mathbf{u})) d^2x dt \\ &= \int_0^T \int_D D \partial_t \phi - \frac{d^2 |\nabla D|^2}{2 D^2} D - \frac{D^2 - 2DD_0}{2\mathcal{F}} \\ &\quad + \frac{1}{2} D |\mathbf{u}|^2 + \mathbf{u} \cdot D \nabla \phi d^2x dt. \end{aligned} \tag{2.1}$$

In (2.2), the term multiplied by d^2 in the action integral S is the time integral of the Fisher–Rao metric. Compute the dynamics for the Fisher–Rao metric ‘information flow’ resulting via Euler–Poincaré equation (1.9) using the Lagrangian in (2.2).

Hint: Upon noticing that the variation of the action integral in (2.2) wrt $\mathbf{u}$ implies $\mathbf{u} = -\nabla\phi$, one could safely apply this constraint and rewrite this Hamilton’s principle for the Legendre transformed Lagrangian as

$$\begin{aligned} S &= \int_0^T \ell(\phi, D) dt = \int_0^T \int_{\mathcal{D}} D \partial_t \phi - \frac{1}{2} D |\nabla \phi|^2 - \frac{d^2}{2} \frac{|\nabla D|^2}{D^2} D - \frac{D^2 - 2DD_0}{2\mathcal{F}} d^2x dt \\ &= \int_0^T \int_{\mathcal{D}} D \partial_t \phi - \tilde{H}(D, \phi) d^2x dt. \end{aligned} \quad (2.2)$$

- (i) Calculate the canonical Hamiltonian equations of motion for $\tilde{H}(D, \phi)$.
- (ii) Upon noticing that $|\nabla D|^2/D = 4|\nabla\sqrt{D}|^2$ set $d^2 = 1$ and transform to polar complex variables $\psi = \sqrt{D}e^{i\phi}$ as done in [Madelung1927].

★

Answer.

- (i) The canonical Hamiltonian equations of motion for $\tilde{H}(D, \phi)$.

$$\begin{aligned} \partial_t D &= \frac{\delta \tilde{H}}{\delta \phi} = \operatorname{div}(D \nabla \phi) \\ \partial_t \phi &= -\frac{\delta \tilde{H}}{\delta D} = -\frac{1}{2} |\nabla \phi|^2 - d^2 \Delta \log D - \frac{1}{\mathcal{F}} (D - D_0) \end{aligned} \quad (2.3)$$

Consequently, the fluid velocity $\mathbf{u} = -\nabla\phi$ satisfies the motion equation,

$$\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -d^2 \nabla \Delta \log D - \frac{1}{\mathcal{F}} \nabla (D - D_0),$$

which reduces to the shallow water motion equation when the parameter d^2 vanishes. Likewise, the density D satisfies the continuity equation for conservation of probability density,

$$\partial_t D = -\operatorname{div}(D \mathbf{u}),$$

which implies conservation of volume for the case of shallow water.

- (ii) Upon making Madelung’s 1927 substitution, $\psi = \sqrt{D}e^{i\phi}$, the Lagrangian becomes

$$\begin{aligned} S &= \int_0^T \tilde{\ell}(\psi, \psi^*) dt = \int_0^T \int_{\mathcal{D}} D \partial_t \phi - \frac{1}{2} D |\nabla \phi|^2 - \frac{1}{2} |\nabla \sqrt{D}|^2 - \frac{D^2 - 2DD_0}{2\mathcal{F}} d^2x dt \\ &= \int_0^T \int_{\mathcal{D}} \Im(\psi^* \partial_t \psi) - \frac{1}{2} |\nabla \psi|^2 - \frac{1}{2\mathcal{F}} (|\psi|^4 - 2|\psi|^2 D_0) d^2x dt \\ &=: \int_0^T \int_{\mathcal{D}} \Im(\psi^* \partial_t \psi) - H(\psi, \psi^*) d^2x dt. \end{aligned} \quad (2.4)$$

Thus, the Madelung change of variables transforms the little-known Hamilton's principle for the Fisher–Rao metric to that for the well-known 2D nonlinear focusing Schrödinger (NLS) equation, given by

$$i\partial_t\psi = -\Delta\psi + \frac{2}{\mathcal{F}}(D_0(\mathbf{x})\psi - |\psi|^2\psi). \quad (2.5)$$

Note that the dependence of $D_0(\mathbf{x})$ on spatial coordinates models bathymetry as a type of potential energy for the evolution of the probability amplitude ψ .

The NLS equation is may be used in studying the probability of creating rogue waves interaction with bathymetry, see [\[\[YMCZW2025\]\]](#).

▲

Numerical Simulations of rogue wave creation by bathymetry

The goals of group project 2.1 are to:

(1) Investigate the role of bathymetry in the creation of rogue waves by making computational simulations of the NLS equation (2.5). This can be accomplished by applying the Fourier spectral method with 1024 and 2048 Fourier nodes in both the x and y directions for a variety of 2D bathymetry functions $D_0(x, y)$ of your choice. First reproduce some of the results to which [\[\[YMCZW2025\]\]](#) refers for NLS models of rogue waves to check your results with flat bathymetry, for example.

(2) Introduce buoyancy into the original model with the Fisher–Rao model and follow the calculations to eventually see how buoyancy enters the NLS equation. Then use your derived result to determine how the combination of bathymetry and horizontal gradients of buoyancy influence NLS wave behaviour by modifying your first code to include buoyancy.

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2.2 Fisher–Rao metric in time – more information geometry

The Fisher–Rao metric for the partial time derivation of the density D in the space of densities $\text{Den}(M)$ defined on a two dimensional manifold $M \subset \mathbb{R}^2$ may be written as

$$\int_{\mathcal{D}} \frac{|\partial_t D|^2}{D^2} D dx dy = 4 \int_{\mathcal{D}} \left(\sqrt{\partial_t D} \right)^2 dx dy \quad (2.6)$$

The Fisher–Rao metric arose when the Lagrangian for a certain set multilayer fluid equations derived in [CHP2010] was restricted to a single layer, as

$$\ell = \frac{1}{2} \int D |\mathbf{u}|^2 + \frac{4d^2}{3} \left(\partial_t \sqrt{D} \right)^2 - g(D - b(\mathbf{x}))^2 dx dy. \quad (2.7)$$

The corresponding Euler–Poincaré single-layer fluid motion equation derived from the Euler–Poincaré variational principle is given by

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} = -g \nabla(D - b) - \frac{d^2}{6} \nabla \left(\frac{1}{\sqrt{D}} \frac{\partial^2 \sqrt{D}}{\partial t^2} \right) \quad (2.8)$$

The last term in (2.8) can be derived quickly by taking the variational steps shown in (1.5),

$$\begin{aligned} \delta \int_0^T \int_{\mathcal{D}} \frac{1}{2} (\partial_t \sqrt{D})^2 d^2 x dt &= \int_0^T \int_{\mathcal{D}} \frac{1}{2} (\partial_t \sqrt{D}) \partial_t \delta \sqrt{D} d^2 x dt = - \int_0^T \int_{\mathcal{D}} (\partial_t^2 \sqrt{D}) \delta \sqrt{D} d^2 x dt \\ &= \int_0^T \int_{\mathcal{D}} \left\langle \frac{-1}{2\sqrt{D}} (\partial_t^2 \sqrt{D}), \delta D \right\rangle dt = \int_0^T \left\langle \frac{-1}{2\sqrt{D}} (\partial_t^2 \sqrt{D}), -\text{div}(D \mathbf{w}) \right\rangle dt \\ &= \int_0^T \left[D \nabla \frac{-1}{2\sqrt{D}} (\partial_t^2 \sqrt{D}) \cdot \mathbf{w} \right] d^2 x dt \quad \text{for any vector field } w = \mathbf{w} \cdot \nabla. \end{aligned}$$

What to do? The travelling wave solution of equation (2.8) for the wave elevation in the case of constant bathymetry $b(\mathbf{x}) = b$ for this system in 1D has the classic *sech*²($x - ct$) form. This makes equation (2.8) very intriguing. A series of lock-release simulations was performed in [CHP2010]. However, much more remains to be done with this class of equations, especially from the viewpoint of transport of probability distributions. We begin by simplifying.

A hierarchy of simplifications of equation (2.8). A hierarchy of equations for describing wave-current interaction (WCI) in nonlinear dispersive wave dynamics on free surfaces was derived in [HH2022] via a sequence of approximations of the Lagrangian in Hamilton’s principle in (2.7) for dispersive nonlinear gravity waves. In particular, replacing D by a constant $\mathfrak{d}$ and smoothing the potential energy in their action integral led to the following Lagrangian.

$$\begin{aligned} \ell_{ModCH2}(\mathbf{u}, D) &= \frac{1}{2} \int d \left(|\mathbf{u}|^2 + \frac{\mathfrak{d}^2}{12} (\text{div } \mathbf{u})^2 \right) \\ &\quad - g \left[(D - b(\mathbf{x})) G_{Q_{op}(d)} * (D - b(\mathbf{x})) \right] d^2 x \end{aligned} \quad (2.9)$$

in which convolution $(*)$ with the Green function $G_{Q_{op}(d)}$ of the Helmholtz operator $(1 - \alpha^2 \Delta)$ acts as a smoothing operator in the potential energy term. The $(\text{div } \mathbf{u})^2$ term in (2.9) effectively replaces the vertical kinetic energy by the divergence of the horizontal velocity.

With one final approximation of the kinetic energy norm to include the full Hilbert norm of the velocity $\|\mathbf{u}\|_{H^1}^2$, produced the Lagrangian for the 2D modified CH2 system which had been studied in 1D in [HOT2009]

$$\begin{aligned} \ell_{H1ModCH2}(\mathbf{u}, D) &= \frac{1}{2} \int d \left(|\mathbf{u}|^2 + \alpha^2 |\nabla \mathbf{u}|^2 \right) \\ &\quad - Fr^{-2} \left[(D - b(\mathbf{x})) G_{Q_{op}^{H1}(\alpha)} * (D - b(\mathbf{x})) \right] d^2 x. \end{aligned} \quad (2.10)$$

Remarkably, the solutions of the equations arising from the Euler–Poincaré variational principle with Lagrangian $\ell_{H1ModCH2}$ in equation (2.10) admits emergent singular solutions for a variety of initial conditions. These singular solutions are identified with the singular momentum map shown in figure 4 for left action of the diffeomorphisms on a semidirect-product Lie algebra. This semidirect-product Lie algebra comprises vector fields representing horizontal current velocity acting on scalar functions representing wave elevation.

The authors in [HH2022] used computational simulations to demonstrate the dynamical interactions of the emergent wavefront trains which are admitted by this special subclass of Lagrangians for a variety of initial conditions. The simulations in [HH2022] applied a Fourier spectral method with 2048 Fourier nodes in both the x and y directions.

In particular, [HH2022] investigated:

- (1) A variety of localised initial current configurations in still water whose subsequent propagation generates surface-elevation dynamics on an initially flat surface; and
- (2) The release of initially confined configurations of surface elevation in still water that generate dynamically interacting fronts of localised currents and wave trains.

The results of these simulations in figures 9 and 10 showed intricate wave-current interaction patterns whose structures are similar to those seen, for example, in Synthetic Aperture Radar (SAR) images taken from the space shuttle shown in figure 8. These wave-current interaction patterns comprise the signature of internal waves on or just below the thermocline of the ocean.

Experience in computationally simulating such patterns represents an essentially unique capability which could be useful in a number of future academic and institutional roles in professional life.

Numerical Simulations of dispersive wave solutions

The goal of group project 2.2 is to report your progress in simulating, analysing and comparing the dispersive wave solutions for lock release (aka dam-break) initial conditions for the Euler–Poincaré motion equations arising from the Lagrangians in (2.7), (2.9) and (2.10) by applying the Fourier spectral method with 1024 and 2048 Fourier nodes in both the x and y directions with both flat and sinusoidal bathymetry.

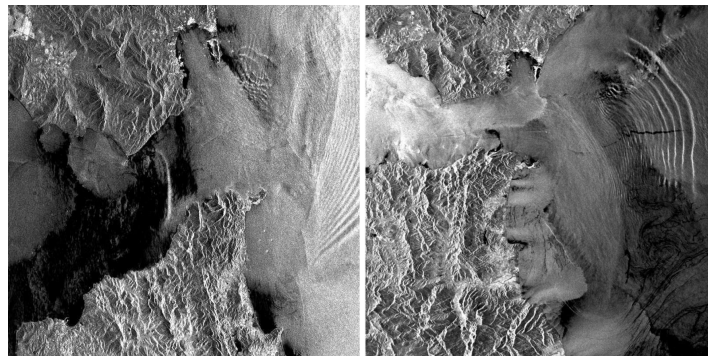


Figure 8: Figure 1: Synthetic Aperture Radar (SAR) image of internal-wave signatures of two snapshots of wave elevation driven by tidal interactions with Atlantic Ocean bathymetry near the Gibraltar Strait. Courtesy of https://earth.esa.int/web/guest/missions/esa-operational-eo-missions/ers/instruments/sar/applications/tropical/-/asset_publisher/tZ7pAG6SCnM8/content/oceanic-internal-waves

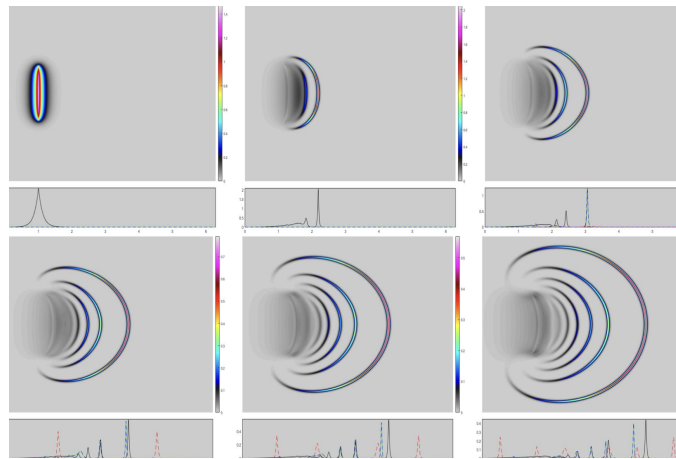


Figure 9: Evolution of wave phase velocity for an initial condition similar to that in Figure 8.

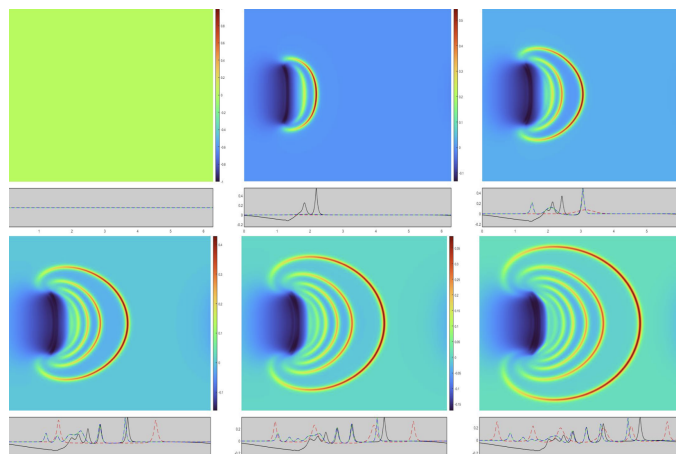


Figure 10: Evolution of wave elevation for an initial condition similar to that in Figure 8.

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2.3 Modified CH2 shallow water system in two dimensions

Consider the Modified CH2 shallow water system in two dimensions (2D) with Lagrangian

$$\begin{aligned} L(\mathbf{u}, \bar{\rho}) = & \frac{1}{2} \int_{\mathcal{D}} \mathbf{u} \cdot (1 - \alpha^2 \Delta) \mathbf{u} \, d^2 \mathbf{x} \\ & + \frac{g}{2} \int (\bar{\rho} - \bar{\rho}_0) (1 - \alpha^2 \Delta) (\bar{\rho} - \bar{\rho}_0) \, d^2 \mathbf{x} \end{aligned} \quad (2.11)$$

where Δ is the 2-dimensional Laplacian operator and $\bar{\rho}_0$ is the constant mean depth. The constant $g > 0$ is the downward acceleration of gravity in applications to shallow water waves. Boundary conditions are taken as $u \rightarrow 0$ and $\rho \rightarrow \rho_0 = \text{const}$ as $|x| \rightarrow \infty$.

The semidirect-product Euler-Poincaré equations for this Lagrangian generalise the modified CH2 system from 1D to 2D dimensions [HOT2009].

2.3.1 Existence of singular solutions

One evaluates the variational derivatives of the Lagrangian in (2.11) as

$$\frac{\delta L}{\delta \mathbf{u}} = \mathbf{m} \cdot d\mathbf{x} \otimes dV \quad \text{and} \quad \frac{\delta L}{\delta \rho} = g\bar{\rho} \otimes dV, \quad (2.12)$$

with covector $\mathbf{m}$ and scalar function ρ defined by

$$\mathbf{m} := (1 - \alpha^2 \Delta) \mathbf{u} \quad \text{and} \quad \rho = (1 - \alpha^2 \Delta) \bar{\rho}. \quad (2.13)$$

Then the semidirect-product Euler-Poincaré equations in $\mathbb{R}^2$ may be written in coordinates as

$$\begin{aligned} \frac{\partial \mathbf{m}}{\partial t} = & - \underbrace{\mathbf{u} \cdot \nabla \mathbf{m}}_{\text{convection}} - \underbrace{(\nabla \mathbf{u})^T \cdot \mathbf{m}}_{\text{stretching}} - \underbrace{\mathbf{m}(\text{div } \mathbf{u})}_{\text{expansion}} - \underbrace{g\rho \nabla \bar{\rho}}_{\text{force}}, \\ \frac{\partial \rho}{\partial t} = & - \text{div } \rho \mathbf{u} \quad (\text{continuity equation}). \end{aligned} \quad (2.14)$$

Here one denotes $(\nabla \mathbf{u})^T \cdot \mathbf{m} = \Sigma_j m_j \nabla u^j$ in components. To explain the terms in underbraces, we rewrite equation (2.14) as the change in the 1-form density of momentum along the characteristic curves of the velocity. In vector coordinates, this is

$$\frac{d}{dt}(\mathbf{m} \cdot d\mathbf{x} \otimes dV) = -g\rho d\bar{\rho} \otimes dV \quad \text{along} \quad \frac{d\mathbf{x}}{dt} = \mathbf{u} = K * \mathbf{m} \quad (2.15)$$

This form of the EPDiff equation also emphasises its nonlocality, since the velocity $\mathbf{u}$ and average depth $\bar{\rho}$ are obtained from the momentum density and the pointwise depth ρ by convolution $K*$ against the Green's function K of the Helmholtz operator.

It is shown in [HOT2009] that in 1D the equation system in (2.14) admits emergent singular solutions in which the momentum density and the depth localise into delta functions. Figure 11 shows an example of a Dam Break initial condition in 1D.

DH: The goal of MFC group project 2.3 is to make progress in determining whether emergent singular solutions can also arise in 2D and demonstrate their emergence in computational simulations. To regularise these simulations, a smoothed version of equations (2.14) which admit emergent singular solutions has been introduced in equations (2.22) and (2.23). Apply the Fourier spectral method with 1024 and 2048 Fourier nodes in both the x and y directions with both flat and sinusoidal bathymetry.

2.3.2 Deriving smoothed versions of the equation set (2.14)

Studying the emergence of singular solutions is computationally problematic. Fortunately, it is possible to apply smoothing operators to recast the equations (2.13) and (2.14) into a computationally tractable form. The more tractable equations are derived next.

In the case of $\mathbb{R}^2$ or the flat 2-torus $\mathbb{T}^2$ is taken for simplicity. There, the flow map $\eta(\mathbf{x}_0, t)$ implies the following relation between Lagrangian and Eulerian velocities,

$$\frac{\partial}{\partial t} \eta(\mathbf{x}_0, t) = \mathbf{u}(\eta(\mathbf{x}_0, t), t). \quad (2.16)$$

Consequently, we may compute the Lagrangian acceleration as the second partial time derivative of η by using the chain rule

$$\frac{\partial^2 \eta}{\partial t^2} = \mathbf{u} \cdot \nabla \mathbf{u} + \frac{\partial \mathbf{u}}{\partial t}. \quad (2.17)$$

Therefore, with notation $Q_{\text{op}} := (1 - \alpha^2 \Delta)$ for the Helmholtz operator, we find

$$Q_{\text{op}} \frac{\partial^2 \eta}{\partial t^2} = [Q_{\text{op}}, (\mathbf{u} \cdot \nabla)] \mathbf{u} + \mathbf{u} \cdot \nabla \mathbf{m} + \frac{\partial \mathbf{m}}{\partial t}. \quad (2.18)$$

Calculating the commutator relation in 2D yields

$$[Q_{\text{op}}, (\mathbf{u} \cdot \nabla)] \mathbf{u} = -\alpha^2 \operatorname{div} (\nabla \mathbf{u} \cdot \nabla \mathbf{u} + \nabla \mathbf{u} \cdot \nabla \mathbf{u}^T) + \alpha^2 (\nabla \mathbf{u}) \cdot \nabla \operatorname{div} \mathbf{u}, \quad (2.19)$$

or, in components,

$$([Q_{\text{op}}, (\mathbf{u} \cdot \nabla)] \mathbf{u})_i = -\alpha^2 \partial_k (u_{i,j} u_{j,k} + u_{i,j} u_{k,j}) + \alpha^2 (u_{i,j}) \partial_j \operatorname{div} \mathbf{u}, \quad (2.20)$$

with a sum on repeated indices. Upon substituting the preceding commutator relation, the EPDiff equation (2.14) and the vector calculus identity

$$-\nabla \mathbf{u}^T \cdot \mathbf{m} = \alpha^2 \operatorname{div} (\nabla \mathbf{u}^T \cdot \nabla \mathbf{u}) - \nabla \left(\frac{1}{2} |\mathbf{u}|^2 + \frac{\alpha^2}{2} |\nabla \mathbf{u}|^2 \right), \quad (2.21)$$

then imply the 2D result

$$\begin{aligned} Q_{\text{op}} \frac{\partial^2 \eta}{\partial t^2} &= Q_{\text{op}} \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) \\ &= -\alpha^2 \operatorname{div} \left(\nabla \mathbf{u} \cdot \nabla \mathbf{u} + \nabla \mathbf{u} \cdot \nabla \mathbf{u}^T - \nabla \mathbf{u}^T \cdot \nabla \mathbf{u} - \nabla \mathbf{u}^T (\operatorname{div} \mathbf{u}) + \frac{1}{2} \operatorname{Id} |\nabla \mathbf{u}|^2 \right) \\ &\quad - \mathbf{u} (\operatorname{div} \mathbf{u}) - \frac{1}{2} \nabla |\mathbf{u}|^2 - g(\bar{\varrho} - \alpha^2 \Delta \bar{\varrho}) \nabla \bar{\varrho}. \end{aligned} \quad (2.22)$$

Similarly, the continuity equation $\partial_t \rho + \operatorname{div} \rho \mathbf{u} = 0$ with $\rho = Q_{\text{op}} \bar{\varrho}$ implies

$$\begin{aligned} Q_{\text{op}} (\partial_t \bar{\varrho} + \operatorname{div}(\bar{\varrho} \mathbf{u})) &= -\alpha^2 \partial_j ((\bar{\varrho} u_{,k}^j)_{,k} + u_{,k}^j \bar{\varrho}_{,k}) \\ &= -\alpha^2 \operatorname{div} (\bar{\varrho} \Delta \mathbf{u} + 2(\nabla \bar{\varrho} \cdot \nabla) \mathbf{u}). \end{aligned} \quad (2.23)$$

Upon applying $K* = Q_{\text{op}}^{-1}$ to smooth both sides of these equations, only spatial derivatives of order $O(1)$ remain and no singularities will arise in the variables $\bar{\mathbf{u}}$ and $\bar{\varrho}$ in equations (2.22). Equations (2.22) and (2.23) comprise a Large Eddy Simulation (LES) model of turbulence in shallow water which has never been studied before.

In 1D equations (2.22) and (2.23) reduce as follows, cf. [HOT2009]

$$\begin{aligned} Q_{\text{op}} (\partial_t u + u u_x) &= -\partial_x \left(u^2 + \frac{\alpha^2}{2} u_x^2 - g(\bar{\varrho} - \alpha^2 \bar{\varrho}_{xx}) \bar{\varrho}_x \right), \\ Q_{\text{op}} (\partial_t \bar{\varrho} + (\bar{\varrho} u)_x) &= -\alpha^2 \partial_x (\bar{\varrho} u_{xx} + 2 \bar{\varrho}_x u_x). \end{aligned} \quad (2.24)$$

Numerical Simulations of the equation set (2.22)

DH: Numerics is where the fun starts!

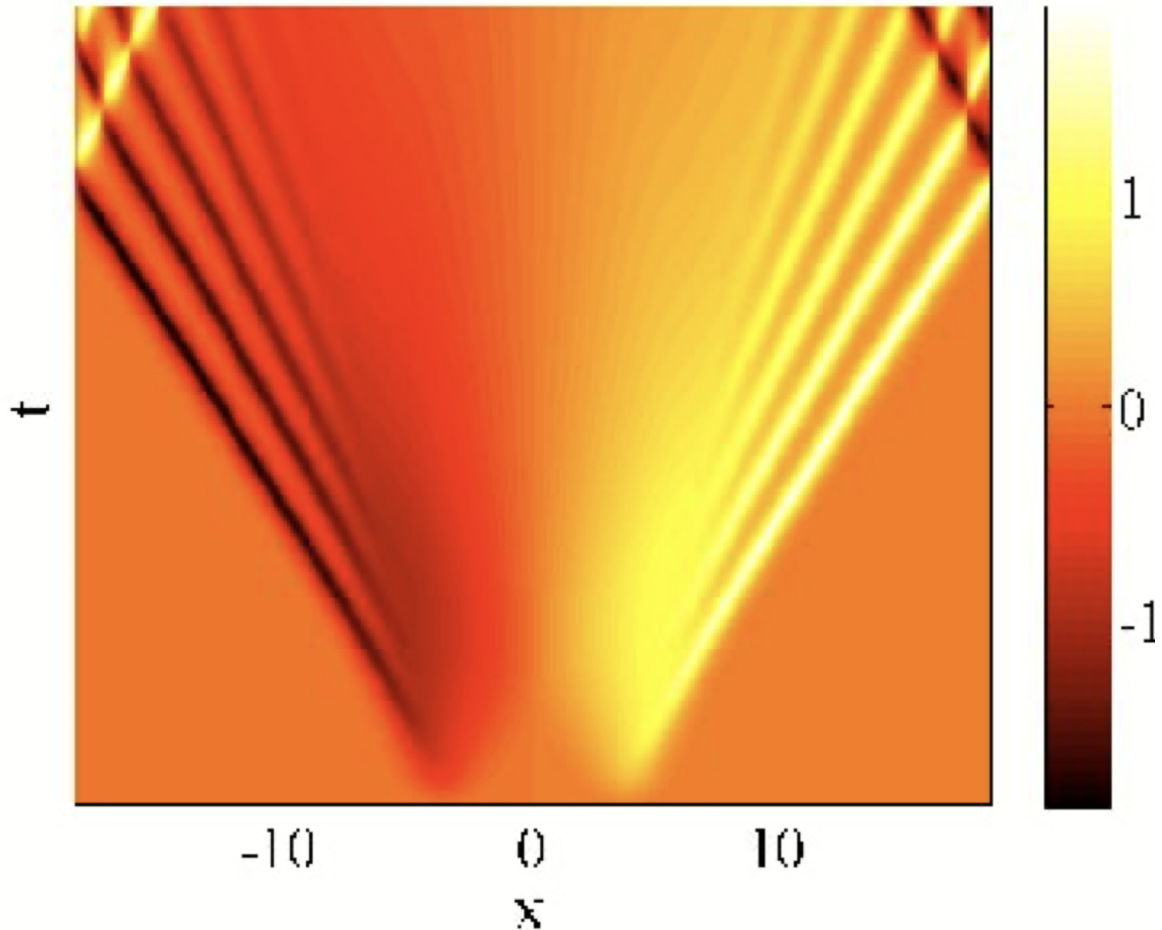


Figure 11: Simulation of a dam-break initial condition for the 1D MCH2 system shows the development of wave trains of smoothed singular solutions propagating in both directions.

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Interlude about modelling wave–current interactions

Wave–current interaction (WCI) dynamics energises and mixes the ocean thermocline by producing a combination of Langmuir circulation, internal waves and turbulent shear flows, which interact over a wide range of time scales.

Two complementary approaches exist for approximating aspects of WCI dynamics.

These are the Generalized Lagrangian Mean (GLM) approach and the Gent–McWilliams (GM) approach.

Their complementarity is evident in their Kelvin circulation theorems. GLM introduces a wave pseudomomentum per unit mass into its Kelvin circulation integrand, while GM introduces a an additional ‘bolus velocity’ to transport its Kelvin circulation loop. The GLM approach models Eulerian momentum, while the GM approach models Lagrangian transport.

In principle, both GLM and GM are based on the Euler–Boussinesq (EB) equations for an incompressible, stratified, rotating flow. The differences in their Kelvin theorems arise from differences in how they model the flow map in the Lagrangian for the Hamilton variational principle underlying the EB equations.

The Euler–Poincaré approach can be used to develop both of these approaches. A framework for doing so would go back the basic Craik–Leibovich (CL) model [CL1976], which provides the foundational reasoning of both GLM and GM. Consequently, we will start with CL.

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2.4 Craik–Leibovich (CL) model → Langmuir turbulence & mixing

Craik–Leibovich: Wind driven sea → Langmuir turbulence → Ocean mix layer [B2012].

In the CL theory, the rapidly oscillating waves at the surface are assumed to be unaffected by the more slowly changing currents below. The effect of the waves on the Eulerian mean flow velocity $\bar{u} := \bar{u}_L - \bar{u}_S$ is parameterized in the CL theory by introducing into the Navier-Stokes equations a “vortex force,” expressed in terms of a *prescribed* Stokes drift velocity, $\bar{u}_S(\mathbf{x})$, which shifts the Lagrangian mean flow velocity $\bar{u}_L$.

The ideal CL equations are given by

$$\partial_t \bar{u} - \bar{u} \times \text{curl} \bar{u} + \nabla \left(\frac{1}{2} |\bar{u}_L|^2 + p \right) = \underbrace{\bar{u}_S \times \text{curl} \bar{u}}_{\text{vortex force}} \quad \text{and} \quad \nabla \cdot \bar{u} = 0. \quad (2.25)$$

These CL equations may be derived from Hamilton’s principle which is similar in appearance to the RSW action integral in (2.58), except the CL equations govern a 3D flow. The constrained Hamilton’s principle for the CL equations is given by

$$S = \int_0^T \int_{\mathcal{D}} \left[\frac{1}{2} D |\bar{u}_L|^2 - D \bar{u}_L \cdot \bar{u}_S - p(D - b(\mathbf{x})) \right] d^3x dt, \quad (2.26)$$

where p is a Lagrange multiplier that enforces incompressibility. Varying the action (2.26) with respect to the Lagrange coordinate $\bar{u}_L$, D and p at fixed x and t gives

$$0 = \delta S = \int_0^T \left\langle D(\bar{u}_L - \bar{u}_S), \delta \bar{u}_L \right\rangle + \langle \mathcal{B}, \delta D \rangle - \langle \delta p, D - 1 \rangle dt, \quad (2.27)$$

In (2.27) the brackets $\langle \cdot, \cdot \rangle$ represent L^2 spatial pairing in 3D and the Bernoulli function has been defined as

$$\mathcal{B} := \frac{1}{2} |\bar{u}_S|^2 - \bar{u}_L \cdot \bar{u}_S - p$$

The variations $\delta \mathbf{u}$ and δD are defined as in equation (1.4), so that

$$\begin{aligned} 0 = \delta S_L &= \delta \int_{\mathcal{D}} \ell(\bar{u}_L, D) dt \\ &= - \int_{\mathcal{D}} \left\langle (\partial_t + \mathcal{L}_{\bar{u}_L}) \frac{\delta \ell}{\delta \bar{u}_L} - D \nabla \frac{\delta \ell}{\delta D}, w \right\rangle dt. \end{aligned} \quad (2.28)$$

The Euler–Poincaré form of Hamilton’s principle in (2.29) and the variational derivatives evaluated in (2.27) imply

$$(\partial_t + \mathcal{L}_{\bar{u}_L}) \left((\bar{u}_L - \bar{u}_S) \cdot d\mathbf{x} \otimes (D d^2x) \right) = d\mathcal{B} \otimes (D d^2x) \quad (2.29)$$

Together, the continuity equation

$$(\partial_t + \mathcal{L}_{\bar{u}_L})(D d^2x) = (\partial_t D + \text{div}(D \bar{u}_L)) d^2x = 0 \quad (2.30)$$

and the constraint $D = 1$ imply $\text{div} \bar{u}_L = 0$. Likewise, the continuity equation and the motion equation together imply

$$(\partial_t + \mathcal{L}_{\bar{u}_L}) \left((\bar{u}_L - \bar{u}_S) \cdot d\mathbf{x} \right) = d\mathcal{B}. \quad (2.31)$$

Writing this equation in terms of $\bar{u} = \bar{u}_L - \bar{u}_S$ yields

$$\partial_t \bar{u} - \bar{u}_L \times \text{curl} \bar{u} + \nabla(\bar{u}_L \cdot \bar{u}) = \nabla \left(\frac{1}{2} |\bar{u}_L|^2 - \bar{u}_L \cdot \bar{u}_S - p \right). \quad (2.32)$$

After a short manipulation, one finds

$$\partial_t \bar{u} - \bar{u} \times \text{curl} \bar{u} + \nabla(|\bar{u}_L|^2 - \bar{u}_L \cdot \bar{u}_S) = \nabla \bar{u}_S \times \text{curl} \bar{u} + \nabla \left(\frac{1}{2} |\bar{u}_L|^2 - \bar{u}_L \cdot \bar{u}_S - p \right), \quad (2.33)$$

which recovers the Craik-Leibovich equation in (2.25),

$$\partial_t \bar{u} - \bar{u} \times \text{curl} \bar{u} + \nabla \left(\frac{1}{2} |\bar{u}_L|^2 + p \right) = \nabla \bar{u}_S \times \text{curl} \bar{u}. \quad (2.34)$$

The corresponding Kelvin theorem partitions into the circulation into a difference $\bar{u} := \bar{u}_L - \bar{u}_S$.

$$\frac{d}{dt} \oint_{\bar{u}_L} \bar{u} \cdot dx = \frac{d}{dt} \oint_{\bar{u}_L} (\bar{u}_L - \bar{u}_S) \cdot dx = - \oint_{\bar{u}_L} dp = 0. \quad (2.35)$$

Notice that the integrand of Kelvin's theorem is Eulerian, while the velocity of its material loop is Lagrangian. This is a fundamental distinction for all fluid dynamics models.⁶

The appearance of the Stokes drift velocity parameterised by $\bar{u}_S$ in the Craik-Leibovich equation tends to produce horizontally oriented vortex cells whose effects on the sea surface near the downwelling flow between vortex cells are seen as 'windrows' accumulating bubbles and other flotsam. These horizontally oriented vortex cells tend to cause vertical mixing known as Langmuir circulation and are thought to be one of the sources of turbulence in the ocean surface boundary layer (OSBL) [B2012].

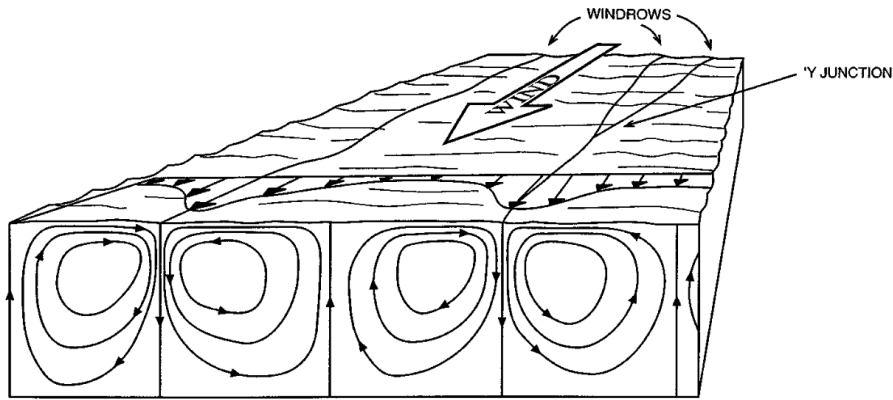


Figure 12: Sketch showing the pattern of mean flow in idealized Langmuir circulation. The windrows may be 2 m to 300 m apart, and the cell form is roughly square (as shown). In practice the flow is turbulent, especially near the water surface, and the windrows amalgamate and meander in space and time. Bands of bubbles or buoyant algae may form within the downward-going (or downwelling) flow [Thorpe2004a].

The goal for group project 2.4 is:

- (1) Transform [H1996] into the Euler–Poincaré form and use the result to discuss Langmuir turbulence.**
- (2) Show that the CL equations follow from an Euler–Poincaré variational principle**
- (3) Distinguish among CL, GLM and GM in terms of their Kelvin circulation theorems.**

⁶Kaluza-Klein form [HMR1998a].

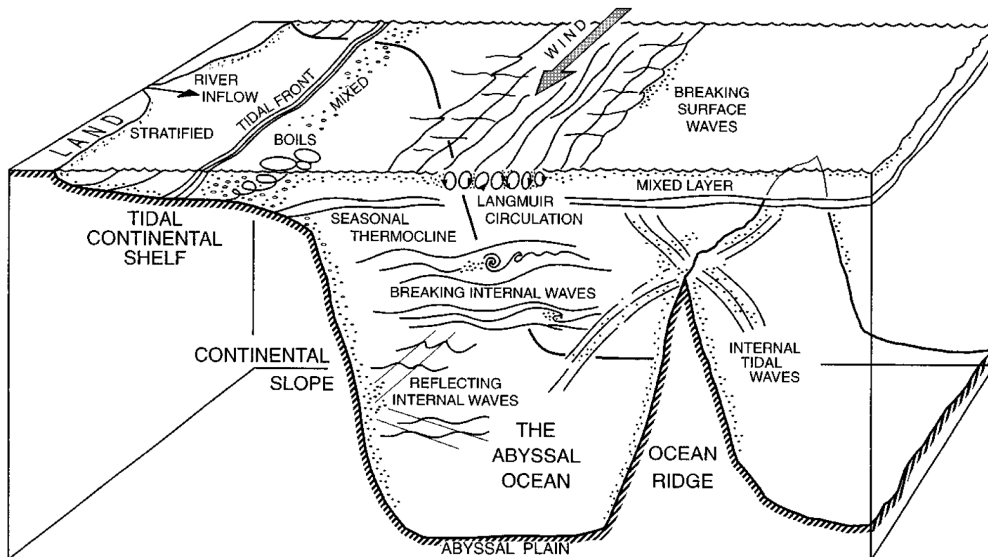


Figure 13: Sources of ocean turbulence [Thorpe2004b].

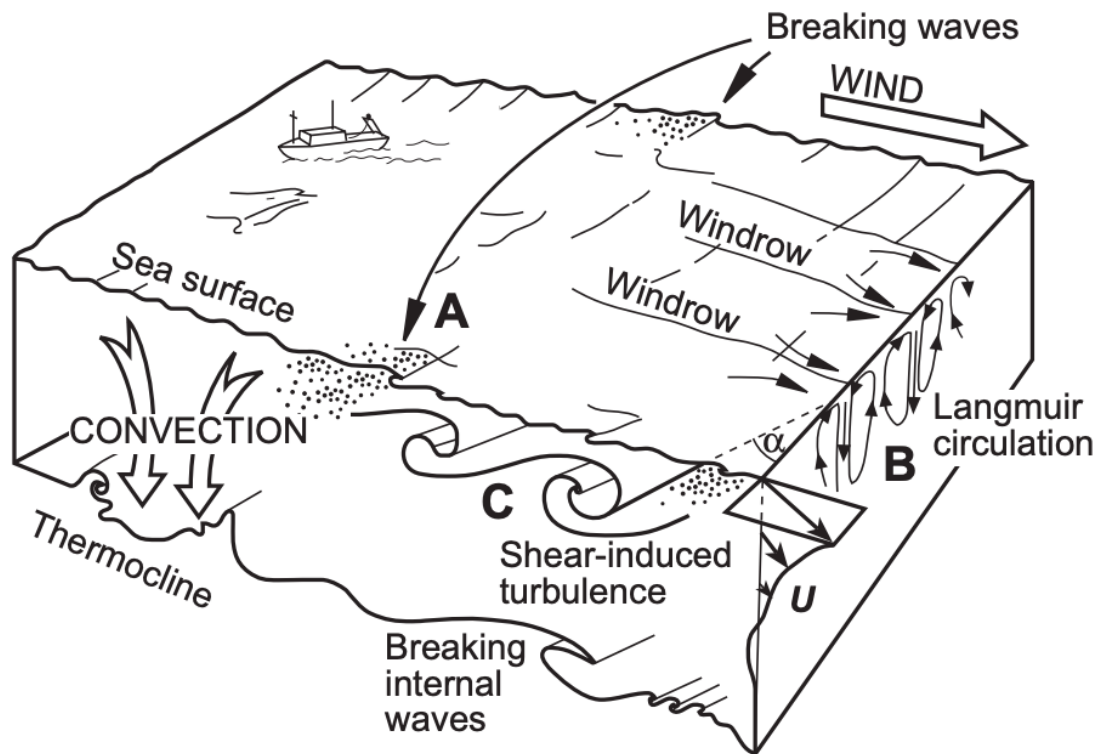


Figure 14: A panoply of air-sea wave-current interactions [Thorpe1995].

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2.5 Generalised Lagrangian Mean (GLM)

GLM is a commonly used approaches for modelling Wave-Current Interaction Dynamics (WCID). Modern derivations and comparison with the Gent–McWilliams parameterisation (GMP) are given in [GV2018, H2019, HHS2023a, HHS2023b].

See <https://www.sciencedirect.com/topics/earth-and-planetary-sciences>

In WCI, the waves are propagating through the moving fluid at a speed comparable to the fluid velocity itself. This means that the wave frequency is Doppler shifted by the fluid motion. However, the wave interaction is by no means frozen into the fluid motion. Instead, the wave-current interaction (WCI) is distributed along the path of the wave through the comparably moving fluid. In particular, the Eulerian mean group velocity of the wave is defined relative to the frame moving with fluid [32], and the Eulerian-mean WCI dynamics at a given fixed point in space depends on the history of wave interaction all along the entire Lagrangian path of the fluid parcel currently occupying that point. Mathematically, this implies that the description of WCI must be formulated in terms of the Eulerian mean of the pull-back of the fluid properties by the Lagrange-to-Euler map, which is assumed to be a smooth invertible map. This is a hybrid description in which the wave activity takes place in the frame of motion of the fluid.

The WCI situation is addressed directly by the Generalized Lagrangian Mean (GLM) approach formulated in Andrews & McIntyre [AM1978a, AM1978a]. GLM generalizes the CL approach by decomposing the flow into its fast and slow components, then taking various types of time-averages, phase-averages and asymptotic approximations of the wave-current interaction dynamics at which the Rossby number is order $O(1)$. In GLM, another dynamical variable is introduced, called the pseudomomentum, in addition to the Stokes mean drift velocity appearing in the CL approach. Thus, in GLM, the fast-slow split in time is performed at a single spatial scale. No differences in spatial scale of the waves and mean flow need to be assumed.

2.5.1 Lagrangian mean & Stokes correction in terms of Eulerian mean

The GLM equations are based on defining fluid quantities at a displaced fluctuating position $\mathbf{x}^\xi := \mathbf{x} + \xi(\mathbf{x}, t, t/\epsilon)$ for $\epsilon \ll 1$. In the GLM description, $\bar{\chi}$ denotes the Eulerian mean of a fluid quantity $\chi = \bar{\chi} + \chi'$ while $\bar{\chi}^L$ denotes the Lagrangian mean of the same quantity, defined by

$$\bar{\chi}^L(\mathbf{x}, t) \equiv \overline{\chi^\xi(\mathbf{x}, t, t/\epsilon)}, \quad \text{with} \quad \chi^\xi(\mathbf{x}, t, t/\epsilon) \equiv \chi(\mathbf{x}(t) + \xi(\mathbf{x}(t), t/\epsilon). \quad (2.36)$$

Here $\mathbf{x}^\xi \equiv \mathbf{x}(t) + \xi(\mathbf{x}(t), t/\epsilon)$ is the current position of a Lagrangian fluid trajectory whose mean position is $\mathbf{x}(t)$. Thus, $\xi(\mathbf{x}(t), t/\epsilon)$ with vanishing Eulerian mean $\overline{\xi(\mathbf{x}(t), t/\epsilon)} = 0$ denotes the fluctuating displacement of a Lagrangian particle trajectory about its current mean position, denoted $\mathbf{x}(t)$.

The key formula for the rate of change of the fast-time average of the Kelvin circulation integral is the following in the notation of [AM1978a],

$$\frac{d}{dt} \oint_{\gamma(\mathbf{x}^\xi)} \overline{\mathbf{u}^\xi(\mathbf{x}, t) \cdot d\mathbf{x}^\xi} = \oint_{\gamma(\bar{\mathbf{x}})} (\partial_t + \mathcal{L}_{\bar{\mathbf{u}}^L}) \left(\left(\bar{\mathbf{u}}_i^L + \overline{u_j^\ell \nabla \xi^j} \right) \cdot d\mathbf{x} \right) \quad (2.37)$$

This formula is the basis for the definition of a quantity known as *pseudomomentum* in the GLM theory. The pseudomomentum $\pi(x, t) := \overline{u_j^\ell \nabla \xi^j}$ with fluctuation velocity $\mathbf{u}^\ell$ plays a role in the GLM theory which is similar to the role of the Stokes drift in the Craik–Leibovich theory of Langmuir turbulence.

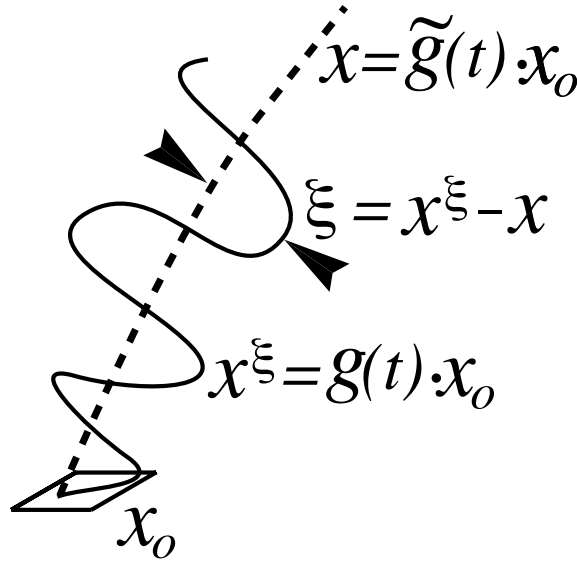


Figure 15: In GLM, oscillating Lagrangian particle trajectories $x^\xi(t) = x(t) + \xi(\mathbf{x}(t), t/\epsilon)$ are averaged over rapid time t/ϵ at each mean time t to create the mean trajectory $\mathbf{x}(t)$.

A motivation for the formula in (2.37) can be offered, by writing the averaged Kelvin circulation, as follows:

$$\begin{aligned} \overline{\oint_{c(\bar{u})} \left(\bar{u} dx + \frac{d\xi(x, t/\epsilon)}{dt} \right) \cdot \left(d(x + \xi(x, t/\epsilon)) \right)} &= \oint_{c(\bar{u})} \left(\bar{u} + \frac{d\xi(x, t/\epsilon)}{dt} \nabla \xi(x, t/\epsilon) \right) \cdot dx \\ &= \oint_{c(\bar{u})} \left(\bar{u}(x, t) + \pi(x, t) \right) \cdot dx. \end{aligned} \quad (2.38)$$

This expression transforms the Kelvin theorem into the Craik–Leibovich form with $\bar{u}_S(x) \rightarrow \pi(x, t)$ with ‘pseudomomentum’ $\pi(x, t)$.⁷

The goal for group project 2.5 is similar to those for 2.4:

- (1) Show that the GLM equations follow from averaging an Euler–Poincaré variational principle.**
- (2) Express GLM theory in Euler–Poincaré form and use the resulting properties to discuss Langmuir turbulence.**
- (3) Distinguish among CL, GLM and GM in terms of their Kelvin circulation theorems.**

⁷Kaluza–Klein form [HMR1998a].

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2.6 Gent–McWilliams parameterisation (GMcP)

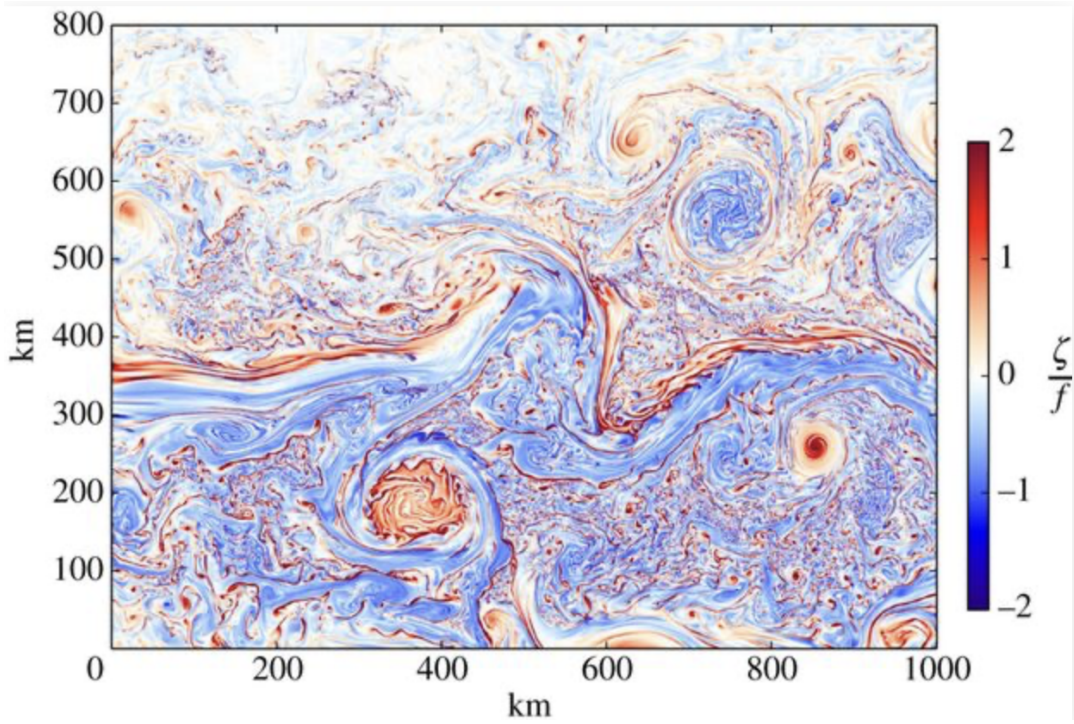


Figure 16: This plot shows a snapshot of vertical vorticity in units of Coriolis frequency arising at the surface in the wintertime Gulf Stream after separation from the western boundary in a nested-subdomain simulation [McW2016]. Note the fine-grained substructure of the ‘eddies’.

In an attempt to enhance heat transport on buoyancy isoclines and thereby slow the vertical diffusion of heat to lower depths, Peter Gent and Jim McWilliams increased the transport velocity of buoyancy by adding a term proportional to the local slope of internal waves on a given buoyancy isocline. This was a direct interference in the buoyancy advection dynamics of *one* of the Primitive Equations. However, the citation rate of the original papers shows that this interference is very popular in computational oceanography, even though it upsets the consistency of the rest of the Primitive Equations.

A shallow water analogue of GMcP. Consider a shallow water analogue of the Gent–McWilliams parameterisation (GMcP) in which the columnar volume of fluid ηd^2x with local depth η moves with a flow velocity given by

$$\mathbf{v} = \mathbf{u} + \kappa \nabla \log \eta \quad (2.39)$$

and the depth η satisfies a continuity equation of the form

$$(\partial_t + \mathcal{L}_v)(\eta d^2x) = 0 = (\partial_t \eta + \operatorname{div}(\eta \mathbf{v})) d^2x. \quad (2.40)$$

Consider a constrained Euler–Poincaré Hamilton principle for this Gent–McWilliams modified shallow water (GMSW) equations in two spatial dimensions, written as

$$\begin{aligned} 0 = \delta S = \delta \int_0^T L(u, \eta) dt &:= \delta \int_0^T l(u, \eta) + \langle p, (\partial_t + \mathcal{L}_v)q \rangle dt \\ \text{with } l(u, \eta) &:= \int_{\mathcal{D}} \frac{1}{2} \eta |u|^2 - \frac{1}{2} g \eta^2 d^2x, \end{aligned} \quad (2.41)$$

in which $\langle \cdot, \cdot \rangle$ denotes L^2 pairing.

Taking variations in the constrained action integral S in (2.41) yields

$$0 = \delta S = \int_0^T \int_{\mathcal{D}} \left\langle \frac{\delta \ell}{\delta u} - p \diamond q, \delta u \right\rangle + \left\langle \frac{\delta \ell}{\delta \eta} + \kappa \eta^{-1} \operatorname{div}(p \diamond q), \delta \eta \right\rangle + \langle \delta p, (\partial_t + \mathcal{L}_v)q \rangle - \langle (\partial_t + \mathcal{L}_v)p, \delta q \rangle dt. \quad (2.42)$$

The corresponding modified Kelvin circulation reveals the effect of the constraint imposed by the momentum $p(\mathbf{x}, t)$ conjugate to the Eulerian quantity $q(\mathbf{x}, t)$ ⁸ as

$$\frac{d}{dt} \oint_{c(v)} \mathbf{u} \cdot d\mathbf{x} = \oint_{c(v)} (\partial_t + \mathcal{L}_v) \mathbf{u} \cdot d\mathbf{x} = \oint_{c(v)} d \frac{\delta L}{\delta \eta} = 0, \quad (2.43)$$

where $c(v)$ is a material loop moving under the action of the transport velocity vector field in (2.39) and

$$\frac{\delta L}{\delta \eta} = \frac{\delta \ell}{\delta \eta} + \frac{\kappa}{\eta} \operatorname{div}(p \diamond q) = \frac{\delta \ell}{\delta \eta} + \frac{\kappa}{\eta} \operatorname{div} \frac{\delta \ell}{\delta u} = \left(\frac{1}{2} |u|^2 - g\eta \right) + \frac{\kappa}{\eta} \operatorname{div}(\eta \mathbf{u}), \quad (2.44)$$

as evaluated by using the expression for $\ell(\mathbf{u}, \eta)$ in (2.41) and the constraint relation in (2.42).

For the group 2.6 project:

(1) Continue the development of the GMcP equations, obtaining its conservation properties, for example;

(2) Develop computational simulations to assess the sensitivity of the solutions on GMcP as a function and sign of the constant parameter κ .

(3) Develop a GMcP model based on modifying buoyancy and depth in TSW (thermal shallow water equations) that would include both depth and buoyancy.

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⁸The Eulerian quantity $q(\mathbf{x}, t)$ tracks the modified Lagrangian trajectory and informs which Lagrangian label is currently at Eulerian position $\mathbf{x}$ under the flow of the modified transport velocity v in (2.39).

2.7 The Lake Equation

Camassa, Holm and Levermore [CHL1996, CHL1997] proposed what they called the ‘lake’ equation, which is a PDE for a planar velocity field $\mathbf{u}(x, y)$ on a domain $\mathcal{D}$,

$$\partial_t \mathbf{u} + (\mathbf{u} \cdot \text{grad}) \mathbf{u} = -\text{grad} p, \quad \text{div}(b\mathbf{u}) = 0, \quad \mathbf{u} \parallel \partial \mathcal{D}, \quad (2.45)$$

where $b(x, y)$ is the bathymetry. The lake equation represents a mean field limit in which the shallow water waves are of such small amplitude compared to the depth (as in an ocean) that the free surface may be regarded as a ‘rigid lid’ against which a local hydrostatic pressure $\hat{h}$ may develop. The hydrostatic rigid lid limit is also imposed in the commonly used ‘primitive equations’ in 3D ocean simulations. In this limit, vertical columns of fluid are assumed to move horizontally in unison, so that the 2D area formula $dA := b(x, y)dx \wedge dy$ representing 3D volume preservation in incompressible flow imposes $\text{div}(b\mathbf{u}) = 0$. Thus the velocity fields take values $u \in \text{SDiff}_\mu(\mathcal{D})$ (smooth invertible maps that preserve measure $\mu = b(x, y)dx \wedge dy$). Well-posedness of the lake equation as a PDE was shown in [LOT1996].

The lake equation in (2.45) may be derived as an Euler–Poincaré equation for Hamilton’s principle whose Lagrangian is similar to the one which generates the 2D Euler equations:

$$\begin{aligned} 0 = \delta S &= \delta \int_0^T \int_{\mathcal{D}} \frac{D}{2} |\mathbf{u}|^2 - \hat{h}(D - b(\mathbf{x})) d^2x dt \\ &= \int_0^T \left\langle \frac{1}{2} |\mathbf{u}|^2 - \hat{h}, \delta D \right\rangle + \langle Du^b, \delta u \rangle - \langle D - b(\mathbf{x}), \delta \hat{h} \rangle dt \\ &= \int_0^T \left\langle \frac{1}{2} |\mathbf{u}|^2 - \hat{h}, -\text{div} D\xi \right\rangle + \langle Du^b, \partial_t \xi - \text{ad}_u \xi \rangle - \langle D - b(\mathbf{x}), \delta \hat{h} \rangle dt \\ &= - \int_0^T \left\langle -D d\left(\frac{1}{2} |\mathbf{u}|^2 - \hat{h}\right) + (\partial_t + \mathcal{L}_u)(Du^b), \xi \right\rangle + \langle D - b(\mathbf{x}), \delta \hat{h} \rangle dt. \end{aligned} \quad (2.46)$$

Thus, upon substituting $u^b = \mathbf{u} \cdot d\mathbf{x}$ and $D - b(\mathbf{x}) = 0$ from the variation $\delta \hat{h}$, we have the lake equation and its constraint in the advection equation for D , as

$$(\partial_t + \mathcal{L}_u)(\mathbf{u} \cdot d\mathbf{x}) - d\left(\frac{1}{2} |\mathbf{u}|^2 - \hat{h}\right) = 0 \quad \text{and} \quad (\partial_t + \mathcal{L}_u)(b(\mathbf{x})d^2x) = \text{div}(b\mathbf{u})d^2x = 0. \quad (2.47)$$

Remark

2.2. Taking the differential of the first of the equations in (2.47), using the property $d^2 = 0$ and defining the lake vorticity ω from the relation $d(\mathbf{u} \cdot d\mathbf{x}) =: \omega dx \wedge dy$ leads by the product rule to

$$(\partial_t + \mathcal{L}_u)\left(\frac{\omega}{b} b d^2x\right) = \left(\frac{\partial}{\partial t} \frac{\omega}{b} + \mathbf{u} \cdot \nabla \frac{\omega}{b}\right) b(\mathbf{x}) d^2x + \frac{\omega}{b} \mathcal{L}_u(b dx \wedge dy), \quad (2.48)$$

with auxiliary equation $\mathcal{L}_u(b(\mathbf{x})dx \wedge dy) = \text{div}(b(\mathbf{x})\mathbf{u})dx \wedge dy = 0$. Hence, one discovers that the ratio ω/b is advected by the lake flow, as

$$\frac{\partial}{\partial t} \frac{\omega}{b} + \mathbf{u} \cdot \nabla \frac{\omega}{b} = 0. \quad (2.49)$$

This advection of the ratio $\omega/b(\mathbf{x})$ implies conservation of the integral quantity

$$C_\Phi := \int_{\mathcal{D}} b \Phi(\omega/b) d^2x, \quad (2.50)$$

for any differential function Φ .

Remark

2.3. Now let ψ be the stream function for the lake equation velocity $\mathbf{u}$, i.e. $\nabla^\perp \psi = b(\mathbf{x})\mathbf{u}$, where $\nabla^\perp$ stands for $(\partial_2, -\partial_1)$. Therefore, one has the following expression for vorticity,

$$\omega = \partial_1 u_2 - \partial_2 u_1 = \partial_1 (-b^{-1} \partial_1 \psi) - \partial_2 (b^{-1} \partial_2 \psi), \quad (2.51)$$

and upon simplifying this equation one obtains the following positive-definite elliptic PDE:

$$-\left(\operatorname{div} \left(\frac{1}{b(x, y)} \nabla \right) \psi = -\left(\frac{\Delta \psi}{b(x, y)} + \nabla \psi \cdot \nabla \left(\frac{1}{b(x, y)} \right) \right) =: \omega. \quad (2.52)$$

If the vorticity were taken to be a delta function, then inverting this elliptic operator to find its Green function would yield the stream function ψ required to determine its point vortex dynamics.

Remark

2.4. Note that the presence of the bathymetry is equivalent to introducing a diagonal metric given by

$$g_{ij}(x, y) = \delta_{ij} b(x, y) \quad (2.53)$$

with determinant $\det g = b^2(x, y)$ so that the area element is

$$dA = \sqrt{\det g} dx \wedge dy = b(x, y) dx \wedge dy \quad (2.54)$$

and the operator $(\operatorname{div} b(x, y)^{-1} \nabla) = \partial_i g^{ij}(x, y) \partial_j$ is the covariant Laplacian for this Riemannian metric.

Thus, the dynamics for point vortex dynamics in the lake equations are the same as the dynamics for point vortex dynamics of the Euler fluid equations on a Riemannian manifold, and that is a well-known branch of mathematics.

The goal of group project 2.7 is to explore the effects of bathymetry in the lake equations, knowing that they are secretly an Euler flow on a Riemannian manifold, whose metric is defined by the physical bathymetry. Find what aspects carry over from the lake equations to the Riemannian Euler equations and vice versa. For example,

- Does the differential geometry of the flow on a Riemannian surface – say, its curvature – have some meaning for the lake flow?
- What are the equations for the dynamics of interacting point vortices for the lake equations? For this part you need the Green function for the positive symmetry operator in equation (2.52).
- What additional mathematics is required to invert the elliptic operator in equation (2.52). Make some progress in simulating these equations.
- Find their equilibrium solutions and the conditions for their stability.
- Derive the *thermal* lake equation and study its properties. In particular, how do horizontal gradients of buoyancy couple to the bathymetry for the thermal lake equation?

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2.8 Stochastic thermal rotating shallow water (TRSW)

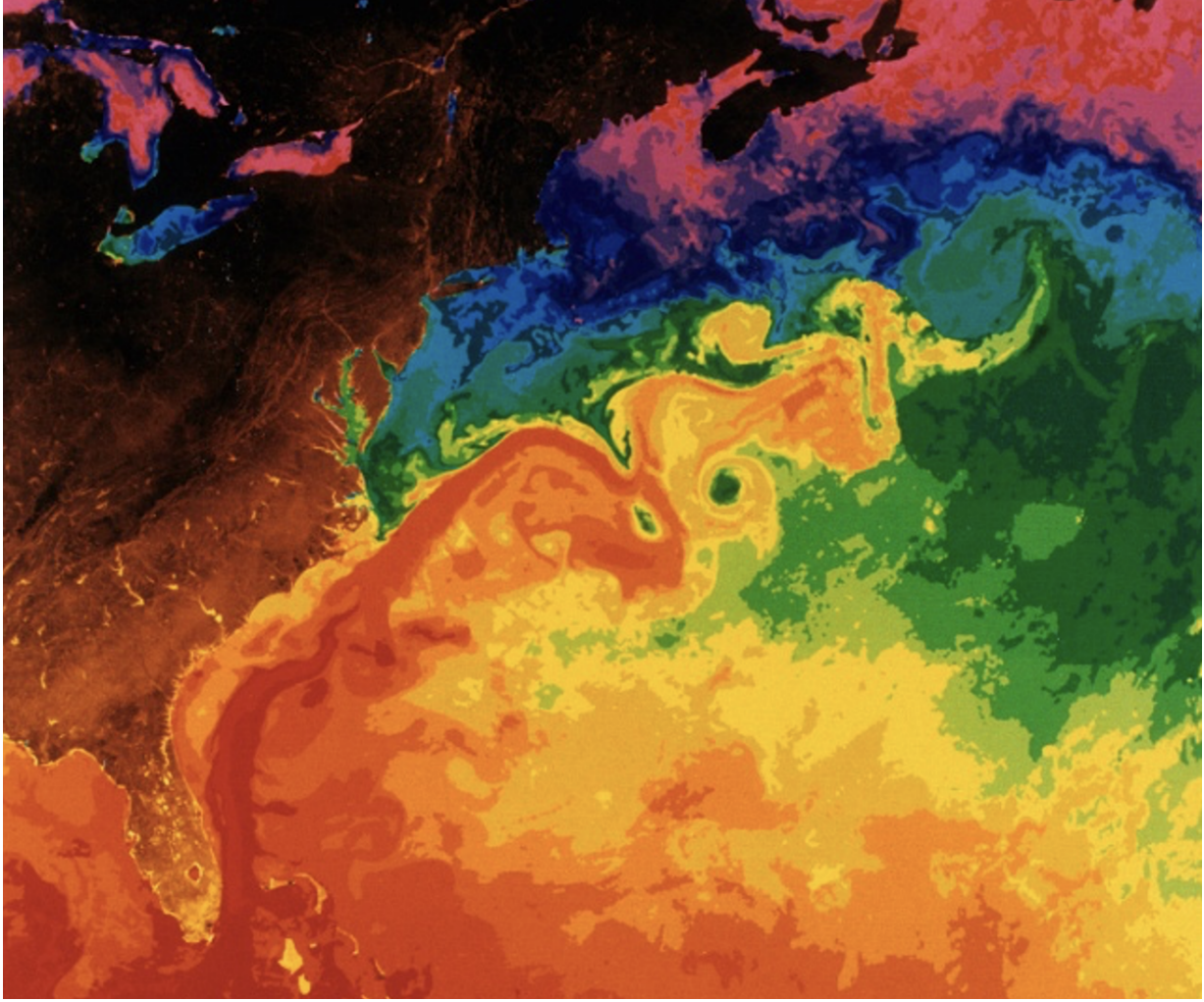


Figure 17: Why is the Gulf Stream getting warmer?

The thermal rotating shallow water (TRSW) model describes an upper active layer of fluid motion with horizontally varying buoyancy moving over an inert lower layer. The TRSW model is an extension of the RSW model and a simplification of the various models we have discussed in the text. This TRSW model comprises an upper active layer of fluid motion with horizontally varying buoyancy and an inert lower layer.

The TRSW equations are expressed using the following definition for the (nonnegative) buoyancy $b(\mathbf{x}, t) = (\bar{\rho} - \rho(\mathbf{x}, t))/\bar{\rho}$, where ρ is the (time and space dependent) mass density of the active upper layer, $\bar{\rho}$ is the uniform mass density of the inert lower layer. We let $\eta = \eta(\mathbf{x}, t)$ be the thickness of the active layer, where $\mathbf{x} = (x, y)$ is the horizontal vector position, and t is time. The nondimensional deterministic TRSW equations are

$$\frac{D}{Dt}\mathbf{u} + \frac{1}{\text{Ro}}f\hat{\mathbf{z}} \times \mathbf{u} + \frac{1}{\text{Fr}^2}\nabla(b\zeta) - \frac{1}{2\text{Fr}^2}(\zeta - h)\nabla b = 0, \quad \frac{\partial \eta}{\partial t} + \nabla \cdot (\eta \mathbf{u}) = 0, \quad \frac{Db}{Dt} = 0, \quad (2.55)$$

with notation Ro for Rossby number and the standard advective time derivative $\frac{D}{Dt} = \partial_t + \mathbf{u} \cdot \nabla$. The boundary conditions in terms of its unit normal vector $\hat{\mathbf{n}}$ are

$$\hat{\mathbf{n}} \cdot \mathbf{u} = 0 \quad \text{and} \quad \hat{\mathbf{n}} \times \nabla b = 0, \quad (2.56)$$

meaning that fluid velocity $\mathbf{u}$ is tangential and buoyancy b is constant on the boundary of the domain of flow.

Upon introducing the following stochastic vector field in $\mathbb{R}^2$ for fluid transport

$$d\chi_t := \mathbf{u}(\mathbf{x}, t)dt + \sum_{i=1}^M \xi_i(\mathbf{x}) \circ dW_t^i, \quad (2.57)$$

we can derive the stochastic TRSW equations. The deterministic equations in (2.55) follow as Euler-Poincaré equations for the action integral

$$S = \int_0^T \ell_{TRSW}(\mathbf{u}, \eta, b) dt = \int_0^T \int_{CS} \left(\frac{1}{2} |\mathbf{u}|^2 + \frac{1}{\text{Ro}} \mathbf{u} \cdot \mathbf{R}(\mathbf{x}) - \frac{1}{2 \text{Fr}^2} b(\eta - 2h) \right) \eta dx dy dt, \quad (2.58)$$

where CS denotes some horizontal surface. The stochastic TRSW equations are derived by first evaluating the variational derivatives for the Lagrangian in the action integral (2.58) as

$$\begin{aligned} \frac{1}{\eta} \frac{\delta l}{\delta \mathbf{u}} &= \mathbf{u} + \frac{1}{\text{Ro}} \mathbf{R}(\mathbf{x}) =: \mathbf{V}(\mathbf{x}, t), \\ \frac{\delta l}{\delta \eta} &= \frac{1}{2} |\mathbf{u}|^2 + \frac{1}{\text{Ro}} \mathbf{u} \cdot \mathbf{R}(\mathbf{x}) - \frac{1}{\text{Fr}^2} b(\eta - h), \\ \frac{\delta l}{\delta b} &= -\frac{1}{\text{Fr}^2} (\eta^2 - 2\eta h). \end{aligned} \quad (2.59)$$

Next, we apply the stochastic Euler-Poincaré theorem with the variational derivatives as above and obtain

$$\begin{aligned} d\mathbf{u} + (d\chi_t \cdot \nabla) \mathbf{u} + (\nabla \xi_i) \cdot \mathbf{u} \circ dW_t^i &= -\frac{1}{\text{Fr}^2} \nabla(b\zeta) dt + \frac{1}{2 \text{Fr}^2} (\zeta - h) \nabla b dt \\ &\quad - \frac{1}{\text{Ro}} f \hat{\mathbf{z}} \times d\chi_t - \frac{1}{\text{Ro}} \nabla(\xi_i \cdot \mathbf{R}) \circ dW_t^i, \\ d\eta + \nabla \cdot (\eta d\chi_t) &= 0, \\ db + (d\chi_t \cdot \nabla) b &= 0. \end{aligned} \quad (2.60)$$

In (2.60), we used $\zeta = \eta - h$ for the free surface elevation.

Remark

2.5. The stochastic Euler-Poincaré equation may be written in three dimensional vector notation as,

$$d\left(\frac{1}{\eta} \frac{\delta l}{\delta \mathbf{u}}\right) - d\chi_t \times \text{curl}\left(\frac{1}{\eta} \frac{\delta l}{\delta \mathbf{u}}\right) + \nabla\left(d\chi_t \cdot \frac{1}{\eta} \frac{\delta l}{\delta \mathbf{u}} - \frac{\delta l}{\delta \eta} dt\right) + \frac{1}{\eta} \frac{\delta l}{\delta b} \nabla b dt = 0. \quad (2.61)$$

For the Lagrangian in (2.58) with variational derivatives given in (2.59) the stochastic Euler-Poincaré equation in (2.61) implies

$$d\mathbf{V} - d\chi_t \times \text{curl} \mathbf{V} + \nabla\left(\mathbf{V} \cdot \xi_i(\mathbf{x}) \circ dW_t^i + \frac{1}{2} |\mathbf{u}|^2 dt\right) + \frac{1}{\text{Fr}^2} \nabla(b\zeta) dt - \frac{1}{2 \text{Fr}^2} (\zeta - h) \nabla b dt = 0. \quad (2.62)$$

Remark

2.6. The stochastic TRSW equations (2.60) imply the following Kelvin circulation law

$$d \oint_{c(d\chi_t)} \frac{1}{\eta} \frac{\delta l}{\delta \mathbf{u}} \cdot d\mathbf{x} = - \oint_{c(d\chi_t)} \frac{1}{\eta} \frac{\delta l}{\delta b} \nabla b \cdot d\mathbf{x}, \quad (2.63)$$

where $c(\mathbf{d}\chi_t)$ is a closed loop moving with stochastic horizontal fluid velocity $\mathbf{d}\chi_t(\mathbf{x}, t)$ in two dimensions. Evaluating for the variational derivatives for TRSW in (2.59) yields

$$\mathbf{d} \oint_{c(\mathbf{d}\chi_t)} \mathbf{V} \cdot d\mathbf{x} = \frac{1}{2 \text{Fr}^2} \oint_{c(\mathbf{d}\chi_t)} (\zeta - h) \nabla b \cdot d\mathbf{x} = \frac{1}{2 \text{Fr}^2} \iint_{\partial S=c(\mathbf{d}\chi_t)} \nabla(\zeta - h) \times \nabla b \, d\mathbf{S} \, dt, \quad (2.64)$$

One sees in equation (2.64) that misalignment of the horizontal gradients of free surface elevation ζ , bathymetry h and buoyancy b will generate circulation.

Remark

2.7. The evolution of potential vorticity on fluid parcels for the TRSW equations in (2.60) is given by

$$\mathbf{d}q + (\mathbf{d}\chi_t \cdot \nabla)q = \frac{1}{2 \text{Fr}^2 \eta} J(\eta, b), \quad (2.65)$$

where the potential vorticity is defined by

$$q := \frac{\varpi}{\eta}, \quad \text{and} \quad \varpi := \hat{\mathbf{z}} \cdot \nabla \times \mathbf{V}, \quad (2.66)$$

and

$$J(\eta, b) = \hat{\mathbf{z}} \cdot \nabla \eta \times \nabla b = -\nabla \cdot (\eta \hat{\mathbf{z}} \times \nabla b) \quad (2.67)$$

is the Jacobian of the depth η .

Remark

2.8. The stochastic TRSW equations (2.60) have an infinite number of conserved integral quantities

$$C_{f,g} = \int_{CS} (f(b) + qg(b)) \eta \, dx dy, \quad (2.68)$$

for the boundary conditions given in (2.56) and any differentiable functions f and g .

Remark

2.9. The Legendre transform which determines the Hamiltonian $\mathbf{d}h$ for the stochastic TRSW equations is defined as⁹

$$\mathbf{d}h(\mu, \eta, b) := \langle \mu, \mathbf{d}\chi_t \rangle - \ell_{\text{TRSW}}(\mathbf{u}, \eta, b) dt, \quad (2.69)$$

in which the angle brackets in the definition of the Legendre transform denote the L^2 pairing over the horizontal cross-section CS . The Hamiltonian form of the stochastic TRSW equations is given by

$$\frac{1}{\text{Fr}} \mathbf{d}F = \left\{ F, \mathbf{d}h \right\} = - \int_{\Omega} \begin{bmatrix} \delta F / \delta \mu_j \\ \delta F / \delta \eta \\ \delta F / \delta b \end{bmatrix}^T \begin{bmatrix} \mu_j \partial_i + \partial_j \mu_i & \eta \partial_i & -b_{,i} \\ \partial_j \eta & 0 & 0 \\ b_{,j} & 0 & 0 \end{bmatrix} \begin{bmatrix} \delta(\mathbf{d}h) / \delta \mu_j = \mathbf{d}\chi_t^j \\ \delta(\mathbf{d}h) / \delta \eta = -\delta \ell_{\text{TRSW}} / \delta \eta \\ \delta(\mathbf{d}h) / \delta b = -\delta \ell_{\text{TRSW}} / \delta b \end{bmatrix} dx dy. \quad (2.70)$$

⁹Show that the Hamiltonian $\mathbf{d}h$ in (2.69) is a semimartingale.

The conserved integral quantities $C_{f,g}$ defined in (2.68) are Casimirs of the Lie–Poisson bracket in (2.70) which persist when the Hamiltonian is made stochastic. This means that these equations describe stochastic coadjoint motion in function space on level sets of the Casimir functionals $C_{f,g}$. Thus, the SALT introduction of stochasticity into the TRSW equations preserves their Lie–Poisson bracket and thereby preserves their geometric interpretation.

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2.9 Geometric Mechanics of the Vertical Slice Model

Abstract: The goals of this group project are to:

- (i) investigate the dynamics of oceanic frontogenesis by taking advantage of the geometric mechanics underlying the class of Vertical Slice Models (VSMs) of ocean dynamics introduced in [CH2013]; and
- (ii) illustrate the versatility and utility of deterministic variational approaches by deriving and simulating wave-current interaction models that describe the effects of internal waves propagating within a vertical planar slice embedded in a 3D region of constant horizontal gradient of buoyancy in the direction transverse to the vertical plane.

The buoyancy variable in the slice $b_s(x, z, t)$ is related to the potential temperature ϑ_s by $b_s = \frac{g}{\vartheta_0} \vartheta_s$. We assume the buoyancy decomposes to $b_s(x, z, t) = \bar{b}_s(z) + b'_s(x, z, t)$ where $\bar{b}_s(z)$ is the background buoyancy which is linear in z . Upon defining the buoyancy frequency as $N^2 = \partial \bar{b}_s / \partial z$, the VSM equations may be written as [HHS2024]

$$\partial_t \mathbf{u}_S + \mathbf{u}_S \cdot \nabla \mathbf{u}_S = f u_T \hat{x} + b'_s \hat{z} - \nabla p, \quad (2.71)$$

$$\partial_t u_T + \mathbf{u}_S \cdot \nabla u_T = -f \mathbf{u}_S \cdot \hat{x} - s \frac{g}{\vartheta_0} \left(z - \frac{H}{2} \right), \quad (2.72)$$

$$\partial_t b'_s + \mathbf{u}_S \cdot \nabla b'_s + N^2 \mathbf{u}_S \cdot \hat{z} + s \frac{g}{\vartheta_0} u_T = 0, \quad (2.73)$$

$$\nabla \cdot \mathbf{u}_S = 0. \quad (2.74)$$

The numerical method to employ here in solving these VSM equations is the compatible finite element method of [CS2012], which is based on the finite element discretisation for VSMs discussed in [YSCMC2017].

(1) Follow the geometric mechanics derivation of VSM in [CH2013, HHS2024].

(2) Compare VSM with the Craik-Liebovich theory, especially in terms of its Kelvin circulation theorem.

(3) Propose a theoretical mechanism for steep fronts to form with VSM initial conditions.

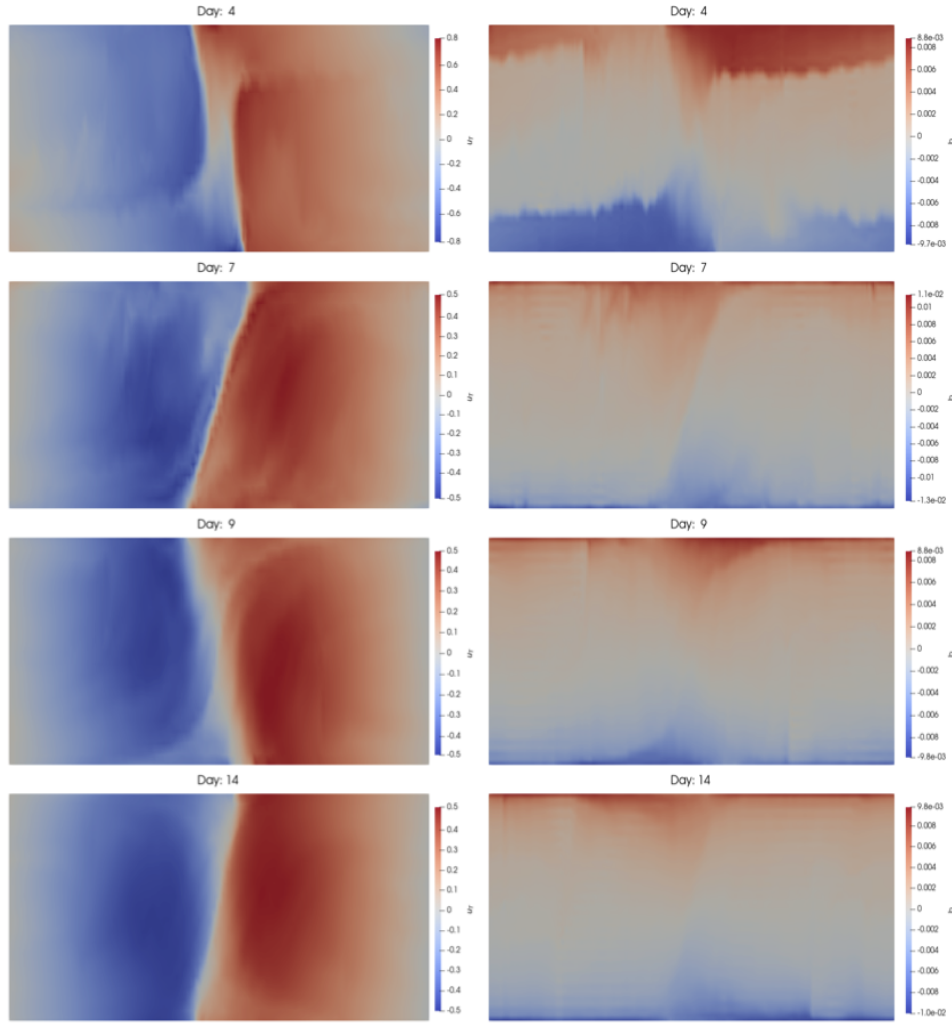


Figure 18: Snapshots of the transverse velocity field u_T (left) and the buoyancy field b'_s (right) of the numerical simulation of the VSM at days 4, 7, 9 and 14 . The solutions are periodic on the lateral boundaries and their in-slice velocity $\mathbf{u}_S$ has no normal component on the top and bottom boundaries; so the buoyancy field is advected horizontally at the top and bottom of the domain.

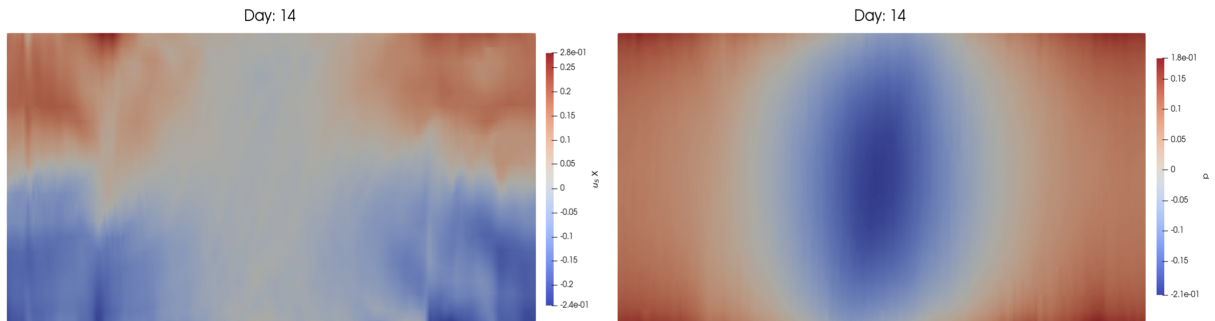


Figure 19: Snapshots of the horizontal component of slice velocity field (left) and the pressure field (right) in the numerical simulation of the VSM equations at day 14.

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2.10 An Euler–Poincaré Hamilton’s principle for quasigeostrophy

An action integral $S = \int L(\mathbf{u}, D) dt$, written in the Eulerian velocity representation with integral operator $\mathcal{G} = (1 - \mathcal{F}\Delta^{-1})$ yields Hamilton’s principle with the constrained Lagrangian

$$L(\mathbf{u}, D) = \int \left(\frac{1}{2} D \mathbf{u} \cdot (1 - \mathcal{F}\Delta^{-1}) \mathbf{u} + D \mathbf{u} \cdot \mathbf{R}(\mathbf{x}) - p(D - 1) \right) dx_1 dx_2$$

and produces the following variations at fixed x_1, x_2 , and t :

$$\begin{aligned} \frac{1}{D} \frac{\delta L}{\delta \mathbf{u}} &= \mathbf{R}(\mathbf{x}) + \mathbf{u} - \frac{\mathcal{F}}{2} \Delta^{-1} \mathbf{u} - \frac{\mathcal{F}}{2D} \Delta^{-1} (D \mathbf{u}), \\ \frac{\delta L}{\delta D} &= \frac{1}{2} \mathbf{u} \cdot (1 - \mathcal{F}\Delta^{-1}) \mathbf{u} + \mathbf{u} \cdot \mathbf{R}(\mathbf{x}) - p, \\ \frac{\delta L}{\delta p} &= 1 - D. \end{aligned}$$

Hence, for action principles of this type a standard vector identity yields the Eulerian QG motion equation,

$$\begin{aligned} \frac{\partial}{\partial t} (1 - \mathcal{F}\Delta^{-1}) \mathbf{u} - \mathbf{u} \times \text{curl} ((1 - \mathcal{F}\Delta^{-1}) \mathbf{u} + \mathbf{R}), \\ + \nabla \left(p + \frac{1}{2} \mathbf{u} \cdot (1 - \mathcal{F}\Delta^{-1}) \mathbf{u} \right) = 0, \end{aligned}$$

where we have used the symmetry of the Laplacian and substituted the constraint $D = 1$ imposed by varying p . When we also take $\text{curl} \mathbf{R}(\mathbf{x}) = f \hat{\mathbf{z}}$ for a constant f , then the curl of the previous equation yields

$$\partial_t \mu + \mathbf{u} \cdot \nabla (\mu + f) + (\mu + f) \nabla \cdot \mathbf{u} = 0 = \partial_t \mu + J(\psi, \mu + f) + (\mu + f) \nabla \cdot \mathbf{u},$$

where $\mu = \hat{\mathbf{z}} \cdot \text{curl} (1 - \mathcal{F}\Delta^{-1}) \mathbf{u}$. The constraint $D = 1$ implies $\nabla \cdot \mathbf{u} = 0$ (from the kinematic relation $\partial D / \partial t + \nabla \cdot D \mathbf{u} = 0$) and when $\mathbf{u} = \hat{\mathbf{z}} \times \nabla \psi$ is substituted, the equation for $\mu = \Delta \psi - \mathcal{F}\psi$ yields the QG potential vorticity equation.

$$\frac{\partial q}{\partial t} + \mathbf{u} \cdot \nabla q = 0, \quad q \equiv \Delta \psi - \mathcal{F}\psi + f. \quad (2.75)$$

For group project 2.9:

(1) The QG version of the 2D Kelvin circulation theorem follows as

$$\frac{d}{dt} \oint_{\gamma(t)} [(1 - \mathcal{F}\Delta^{-1}) \mathbf{u} + \mathbf{R}(\mathbf{x})] \cdot d\mathbf{x} = 0,$$

for any closed loop $\gamma(t)$ moving with the QG fluid velocity $\mathbf{u}$ under the Eulerian motion equation.

(2) Show that the QG motion equation is may be derived in Euler-Poincaré form via the Kaluza-Klein construction [HMR1998a].

(3) For a geodesic stationary principle, the Lagrangian and the Hamiltonian coincide. Show that in our case, Legendre transforming the Lagrangian gives the Hamiltonian,

$$H = \int \left(\frac{1}{2} D \mathbf{u} \cdot (1 - \mathcal{F}\Delta^{-1}) \mathbf{u} + p(D - 1) \right) d^2x$$

so that the Lagrangian and the Hamiltonian do coincide for QG, in the f -plane theory and on the constraint manifold.

(4) Write the QG equations in spherical coordinates and discuss the interpretation of figure 20, courtesy of [LFEG2024]

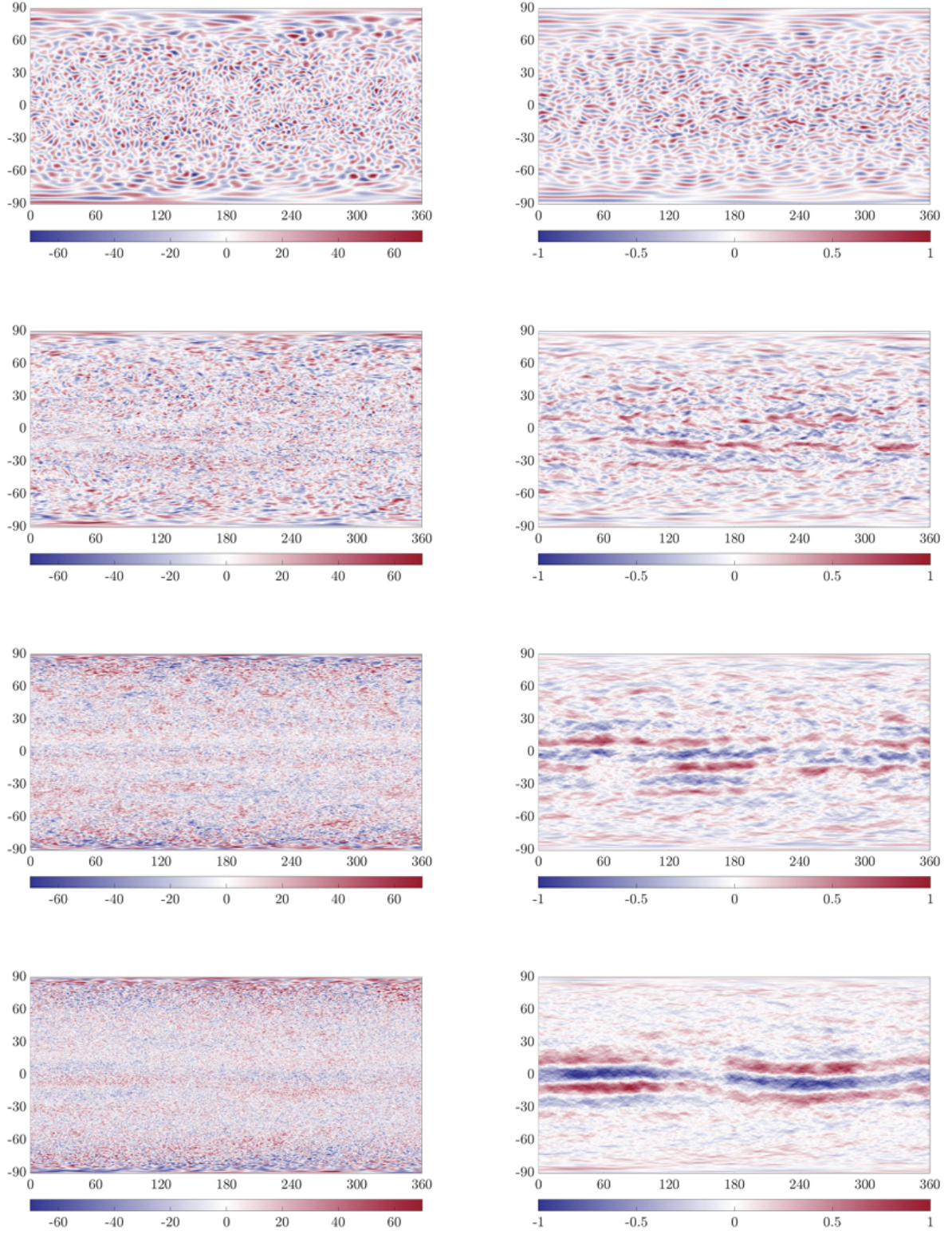


Figure 20: A sequence of solutions of freely evolving quasi-geostrophic flow on the sphere. Shown are the instantaneous potential vorticity anomaly (left column) and zonal component of the horizontal velocity field (right column) at the initial time (top row), after 100 days (second row), after 400 days (third row) and after 2000 days (bottom row).

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